New Wine in an Old Bottle

NeuralPOM: A Hybrid Ocean Forecast Model with Differentiable Dynamics and Staged Learning

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Abstract

Numerical ocean circulation models integrate the three-dimensional primitive equations, but finite grid resolution and empirical subgrid parameterizations limit their forecast skill at eddy-permitting scales. Purely data-driven neural forecasters can advance ocean states at lower computational cost, yet over extended integrations they accumulate trajectory drift, violate volume and tracer conservation, and damp small-scale variability. Here we present NeuralPOM, a hybrid ocean forecasting model that couples differentiable explicit mode-split and semi-implicit free-surface implementations of the Princeton Ocean Model to a conservation-constrained neural corrector. The physical dynamical framework advances the three-dimensional primitive equations across time steps, while the neural network periodically corrects horizontal velocity, potential temperature, and salinity under volume, heat, and salt conservation constraints. We construct a fine-grid reference dataset exceeding 100,000 model days in a wind-driven double-gyre ocean basin and introduce a three-stage training procedure for hybrid ocean models comprising offline single-step pretraining, multi-step differentiable training through the physical dynamical framework, and multi-day closed-loop rollout training. Over 100-day forecasts, NeuralPOM reduces root-mean-square errors across all prognostic variables relative to coarse-grid physical parameterizations and direct neural forecasters, maintaining near-grid kinetic energy while improving mesoscale eddy phase alignment. Stage and controlled route ablations demonstrate the complementary roles of each training stage, showing that backpropagating gradients through the differentiable dynamical framework during multi-step training stabilizes thermohaline and circulation evolution under hourly coupling, while closed-loop rollout training suppresses cumulative trajectory drift over multi-day horizons. Furthermore, physical diagnostics reveal that under high-frequency coupling, uncoordinated temperature and salinity increments project onto density anomalies via the nonlinear equation of state and excite spurious

baroclinic pressure-gradient tendencies. Based on this mechanism, we introduce Density-Normal Damping (DND) to selectively attenuate the density-modifying (diapycnal) component of each layerwise tracer increment while preserving the density-compensated (isopycnal) water-mass correction, improving 100-day forecast skill across all prognostic fields.

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1 Introduction

Forecasting the physical state of the ocean requires predicting the coupled three-dimensional evolution of horizontal and vertical currents, free-surface elevation, potential temperature, salinity, seawater density, and turbulent mixing. Operational and climate-scale ocean models address this problem by numerically integrating the hydrostatic, Boussinesq primitive equations under prescribed atmospheric forcing and basin geometry [1, 2]. At grid spacings affordable for extended ensembles or multi-month integrations, however, numerical models only partially resolve the baroclinic instabilities and mesoscale eddies that govern much of the ocean's kinetic energy and tracer transport [3], while empirical subgrid closures must balance numerical stability against excessive dissipation of resolved scales. Conversely, purely data-driven neural forecasters can advance daily ocean states at lower computational cost than numerical solvers [4–7], yet over multi-week autoregressive integrations they exhibit compounding trajectory drift [4, 5, 8],

progressive attenuation of near-grid mesoscale variability under pointwise mean-squared-error objectives [8–10], and lack built-in volume, heat, and salt conservation constraints across interior layers [10, 11].

A natural question is whether numerical circulation models and machine learning can be combined into a hybrid framework that retains the conservation laws and dynamical balance of a numerical solver while leveraging data-driven corrections for unresolved processes. Recent studies have embedded learned subgrid parameterizations within idealized and regional ocean models [12–17], demonstrated end-to-end differentiable training in atmospheric general circulation models [10], and coupled physics-guided neural correctors and assimilation modules with ocean dynamical frameworks [11, 18–20]. Most ocean hybrid studies to date, however, have focused on idealized quasi-geostrophic, barotropic, or shallow-water configurations, leaving open how neural state updates interact with a complete three-dimensional, terrain-following primitive-equation circulation model that prognostically evolves barotropic and baroclinic velocity modes, a free surface, active temperature and salinity, a nonlinear equation of state (EOS), and a two-equation vertical turbulence closure.

Coupling a neural corrector to a three-dimensional primitive-equation ocean model introduces three physical and numerical challenges that do not arise in simplified models. First, because each correction alters the state trajectory visited by the physical dynamical framework, a network trained only on offline single-step targets experiences distribution shift during closed-loop integration [15, 21], requiring staged training procedures that combine multi-step differentiable training through the dynamical framework with closed-loop rollout training [22–24]. Second, advancing the solver across time steps requires complete multi-time-level state histories, including current and backward leapfrog levels, barotropic–baroclinic mode-coupling transports, and turbulent kinetic energy and mixing-length fields; unconstrained state updates that violate leapfrog time-level consistency or column volume budgets excite computational modes and fast external gravity waves. Third, potential temperature and salinity are active thermodynamic variables that enter the horizontal momentum equations through the nonlinear EOS [25] and hydrostatic baroclinic pressure-gradient force, while modulating vertical viscosity and diffusivity through the buoyancy frequency in the turbulence closure [26]. Consequently, a tracer increment that reduces instantaneous pointwise error at a single step can still introduce small-scale density anomalies, generate spurious baroclinic pressure-gradient tendencies, and perturb vertical mixing over subsequent integration steps.

Here we present NeuralPOM, a hybrid ocean forecasting system built upon the three-dimensional Princeton Ocean Model (POM) [27, 28]. NeuralPOM combines two core components: differentiable PyTorch POM dynamical frameworks—covering both explicit mode-split and semi-implicit free-surface formulations verified against double-precision Fortran POM across idealized and coastal benchmarks—and a residual U-Net corrector that periodically updates horizontal velocity, potential temperature, and salinity under volume, heat, and salt conservation constraints. Using a fine-grid reference ensemble exceeding 100,000 model days in a wind-driven double-gyre ocean basin, we train NeuralPOM through a three-stage training procedure comprising offline single-step pretraining, multi-step differentiable training through the physical dynamical framework, and 10-day closed-loop rollout training. Over 100-day forecasts, NeuralPOM reduces forecast errors across all prognostic variables relative to coarse-grid physical parameterizations and direct neural forecasting architectures [29–32]. Stage progressions and controlled route ablations demonstrate that multi-step differentiable training and closed-loop rollout training play complementary roles: backpropagating gradients through the differentiable dynamical framework during multi-step training stabilizes thermohaline and circulation evolution under hourly coupling, while multi-day closed-loop rollout training suppresses cumulative trajectory drift across both daily and hourly coupling intervals. Furthermore, diagnostic perturbation experiments and 100-day closed-loop integrations reveal that under hourly coupling, uncoordinated temperature and salinity increments project onto density anomalies via the nonlinear EOS and trigger spurious baroclinic pressure-gradient feedback; decomposing each layerwise

tracer increment relative to the local EOS density gradient and applying Density-Normal Damping (DND) to attenuate the density-modifying (diapycnal) component while preserving the density-compensated (isopycnal) water-mass correction suppresses this instability and consistently improves long-range skill. Finally, wavenumber spectral analysis shows that NeuralPOM maintains physical eddy kinetic energy and near-grid variability while reducing spectral error through improved mesoscale phase alignment.

2 The NeuralPOM Framework

NeuralPOM alternates two operations across forecast cycles: a differentiable three-dimensional primitive-equation dynamical framework advances the complete numerical ocean state over a physical integration interval Δ, and a conservation-constrained neural corrector updates the prognostic baroclinic velocity and thermohaline fields at the end of each interval (Fig. 1).

2.1 Differentiable primitive-equation frameworks

Both dynamical frameworks integrate the hydrostatic, Boussinesq primitive equations of the Princeton Ocean Model (POM) on a staggered Arakawa-C horizontal grid with terrain-following σ-coordinates [27]. With horizontal velocity $\boldsymbol{u}_h = (u, v)$, vertical velocity w, material derivative D_t, Coriolis parameter f, reference density ρ_0, hydrostatic pressure p, in-situ density ρ, and prognostic tracers $C \in \{T, S\}$, the continuous governing equations are

$$D_t \boldsymbol{u}_h + f \boldsymbol{e}_z \times \boldsymbol{u}_h = -\rho_0^{-1} \nabla_h p + \mathcal{D}_m, \qquad \partial_z p = -g\rho, \tag{2.1}$$

$$\nabla_h \cdot \boldsymbol{u}_h + \partial_z w = 0, \qquad D_t C = \mathcal{D}_C, \tag{2.2}$$

where $\mathcal{D}_m$ and $\mathcal{D}_C$ represent subgrid momentum and tracer mixing. Seawater density $\rho(T, S, P)$ is evaluated from the nonlinear Mellor equation of state (EOS) [25], vertical viscosity K_M and diffusivity K_H are obtained from the prognostic two-equation Mellor–Yamada level-2.5 turbulence closure for turbulent kinetic energy and length scale $(q^2, q^2 l)$ [26], and reference horizontal dissipation follows the harmonic Smagorinsky formulation ($A_H = 0.2 A_M$).

The free-surface elevation η evolves according to the vertically integrated continuity equation $\partial_t \eta + \nabla_h \cdot (H\boldsymbol{U}) = 0$, where $H = h + \eta$ is the total water-column depth and $\boldsymbol{U} = (U, V)$ is the depth-averaged horizontal velocity. We implemented two differentiable PyTorch dynamical frameworks that share identical three-dimensional baroclinic, thermohaline, and turbulence operators at the internal time step $\Delta t = 600\,\mathrm{s}$ but differ in their numerical integration of fast external gravity waves. In the explicit mode-split framework (*Explicit*), the depth-integrated barotropic mode is subcycled explicitly over $N_e = 30$ leapfrog steps ($\delta t = 20\,\mathrm{s}$) per internal step with Robert–Asselin time filtering, accumulating interval-averaged free-surface elevations and face transports to maintain volume consistency with the three-dimensional baroclinic mode.

In the semi-implicit free-surface framework (*Implicit*), external subcycling is replaced by an elliptic Helmholtz solve for the forward free-surface elevation η^f at $\Delta t = 600\,\mathrm{s}$ using preconditioned conjugate gradients (PCG) with a discrete sine transform preconditioner; during reverse-mode differentiation, a custom implicit-differentiation adjoint solves the area-weighted self-adjoint linear system directly rather than backpropagating through the PCG iterations, accounting for the sensitivity of the elliptic operator to total water depth. Appendix A presents the discrete governing equations and adjoint formulation, and Appendix F verifies both PyTorch frameworks against double-precision Fortran POM across double-gyre, seamount, idealized coastal, and realistic Jiaozhou Bay tidal benchmarks. In both formulations, potential

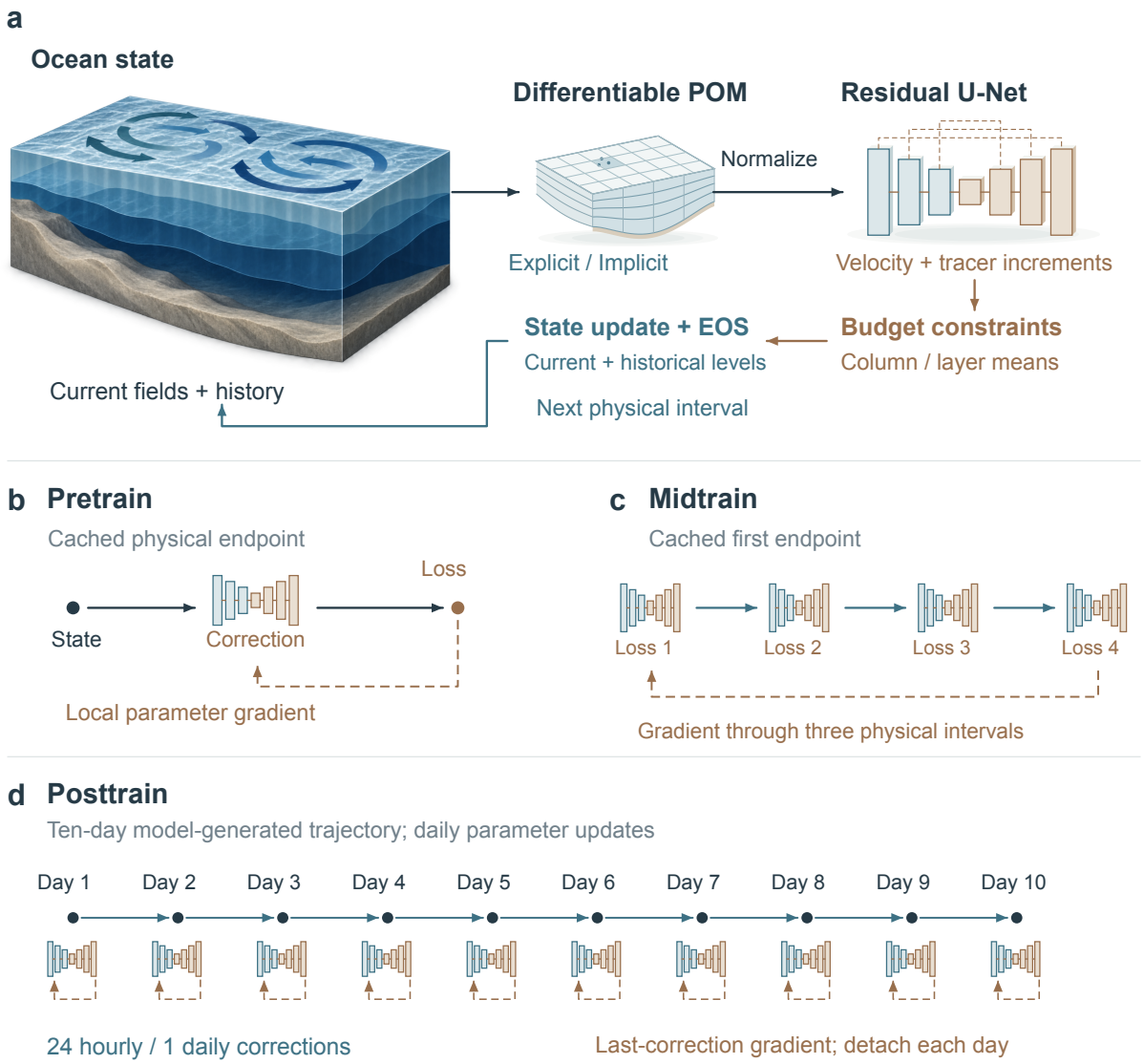


Figure 1. Physical–neural coupling and three-stage training. **a,** At each correction step, the physical framework supplies standardized inputs to the residual U-Net. Velocity and tracer increments satisfy weighted column-mean and layer-mean constraints, respectively, before current and backward leapfrog fields are updated synchronously and density is recomputed from the equation of state (EOS). The ocean rendering is a schematic illustration. **b,** Pretrain optimizes a single correction step on a cached physical forecast. **c,** Midtrain averages losses across four correction steps, retaining three intervening physical integration intervals in the computational graph. **d,** Posttrain integrates 10-day closed-loop trajectories with daily parameter updates. Each day spans 24 hourly corrections or one daily correction; only the final correction of each day contributes to the parameter gradient, and the state is detached between days. Warm brown dashed arrows denote retained parameter-gradient paths.

temperature and salinity force the momentum and turbulence equations through the diagnostic density ρ via hydrostatic baroclinic pressure gradients and vertical buoyancy stratification, and advancing the solver across intervals requires the complete 45-variable state vector across current (n) and backward (b) leapfrog time levels.

2.2 Constrained state updates and density-normal damping

Let x_n^+ denote the complete numerical ocean state after correction cycle n, $\mathcal{P}^\Delta$ the physical integration operator over $\Delta \in \{1\,\mathrm{h}, 1\,\mathrm{day}\}$, and $\mathcal{K}_\theta$ the constrained neural correction operator. Each coupled forecast cycle advances according to

$$x_{n+1}^- = \mathcal{P}^\Delta(x_n^+), \qquad x_{n+1}^+ = \mathcal{K}_\theta(x_{n+1}^-). \tag{2.3}$$

At the end of each physical interval, a residual U-Net maps 101 standardized input channels—comprising $(u^n, u^b, v^n, v^b, T^n, S^n)$ across all 16 vertical storage levels together with η_{E}^n and (U^n, U^b, V^n, V^b)—to 60 raw increment channels $(\delta u, \delta v, \delta T, \delta S)$ across the 15 active σ-layers, leaving the inactive bottom storage level unmodified (Appendix B.2).

To ensure that the neural increments respect the conservation laws and time-stepping structure of the primitive-equation solver, raw network outputs are filtered through a sequence of physical projections before updating the model state (Appendix B.3). Horizontal velocity increments are first projected to zero thickness-weighted column mean ($\sum_{k=0}^{14} \Delta\sigma_k \delta u_{ijk} = 0$ and $\sum_{k=0}^{14} \Delta\sigma_k \delta v_{ijk} = 0$), restricting direct momentum updates to the vertical baroclinic shear so that depth-integrated volume transport remains governed by the external continuity equation. Temperature and salinity increments are similarly projected to zero volume-weighted horizontal mean on every active σ-layer ($\sum_{ij} w_{ijk} \delta T_{ijk} = 0$ and $\sum_{ij} w_{ijk} \delta S_{ijk} = 0$, where $w_{ijk} = A_{ij} m_{ij} H_{\mathrm{B},ij} \Delta\sigma_k$ is the wet cell volume), redistributing heat and salt laterally without introducing net layer sources or sinks.

Within each σ-layer, the standardized layer-mean-preserving tracer increment $z = (\delta T'/s_T, \delta S'/s_S)$ is then decomposed relative to the local EOS density gradient into a density-modifying (diapycnal) projection Ez onto the subspace $\mathcal{S}$ aligned with the local unit density normal and an orthogonal density-compensated (isopycnal) component $(I - E)z$, both of which preserve zero horizontal layer means:

$$z_c = (I - E)z + cEz. \tag{2.4}$$

Setting $c = 1$ yields the *Undamped* tracer update, whereas setting $c = 3/4$ defines *Density-Normal Damping* (DND), which attenuates the density-modifying component by 25% in amplitude (43.75% in squared norm) while retaining the density-compensated water-mass correction in full (Section 5 and Appendix C). Finally, common domain-wide scalar factors cap the standardized velocity pair $(\delta u', \delta v')$ and tracer pair $(\delta T', \delta S')$ at $z_{\max} = 5$ without altering their zero-mean budgets. The resulting physical increments are added identically to both current (n) and backward (b) leapfrog levels ($f^n \leftarrow f^n + \widehat{\delta f}$ and $f^b \leftarrow f^b + \widehat{\delta f}$ for $f \in \{u, v, T, S\}$), shifting the state trajectory while keeping the leapfrog difference $f^n - f^b$ invariant so as not to excite computational modes. Three-dimensional density ρ is immediately recomputed from the updated (T, S) fields via the nonlinear POM EOS, while free-surface elevation η, transformed vertical velocity Ω, turbulence variables $(q^2, q^2 l, K_M, K_H, K_Q)$, and barotropic transport histories adjust dynamically during the subsequent physical integration $\mathcal{P}^\Delta$.

2.3 Physics-informed objective and three-stage training procedure

At each scored correction step, the updated state x is evaluated against the coarse-grained fine-grid reference state y using a composite objective $\ell_\chi(x, y)$ (Appendix B.4) that combines volume-weighted normalized mean squared errors (MSE) in current and backward horizontal velocity ($0.25\,N_m$), potential temperature ($0.25\,N_T$), and salinity ($0.25\,N_S$) with a discrete hydrostatic baroclinic pressure-gradient consistency term ($0.15\,N_p$), a horizontal tracer-gradient term ($0.10\,N_F$), and, during multi-step trajectory training ($\chi = 1$), a free-surface elevation term ($0.10\,N_\eta$).

Training proceeds through three sequential stages designed to transition the corrector from offline single-step initialization to multi-day closed-loop integration (Fig. 1b–d and Appendix B.5). In the first stage, single-step pretraining (*Pretrain*), the final convolution layer is initialized to zero and the network is trained on cached single-interval physical forecasts $v = \mathcal{P}^{\Delta}(x_0)$ by minimizing $J_{\mathrm{pre}}(\theta) = \mathbb{E}\,\ell_0(\mathcal{K}_\theta(v), y)$ with the numerical solver outside the automatic-differentiation graph. In the second stage, multi-step differentiable training (*Midtrain*), the network is initialized from the Pretrain checkpoint and unrolled across four correction cycles connected by three online physical integration intervals retained within the differentiable solver graph (spanning 4 h for $\Delta = 1$ h and 4 days for $\Delta = 1$ day):

$$J_{\mathrm{mid}}(\theta) = \frac{1}{4}\sum_{j=1}^{4} \ell_1(x_j^+, y_j), \qquad x_j^+ = \mathcal{K}_\theta\big(\mathcal{P}^{\Delta}(x_{j-1}^+)\big). \tag{2.5}$$

Backpropagating through $\mathcal{P}^{\Delta}$ allows endpoint errors at steps $j > 1$—including free-surface elevation and velocity errors—to constrain earlier state updates at step $j - 1$ through the linearized primitive equations. In the final stage, closed-loop rollout training (*Posttrain*), the model is initialized from the Midtrain checkpoint and integrated continuously over 10-day closed-loop rollouts; parameters θ are updated at the end of each 24-hour window using the detached pre-correction state v_n,

$$L_{\mathrm{post},n}(\theta) = \ell_1\big(\mathcal{K}_\theta(\mathrm{sg}(v_n)), y_n\big), \tag{2.6}$$

after which the updated state is detached before advancing into day $n+1$ (corresponding to one correction cycle per day for $\Delta = 1$ day, or 23 forward-only hourly cycles followed by one gradient-tracked cycle per day for $\Delta = 1$ h). This stage adapts the corrector to the state distribution encountered during extended closed-loop integrations without requiring backpropagation across 10-day horizons.

3 Double-Gyre Benchmark and Experimental Protocol

3.1 Wind-driven double-gyre reference dataset

We evaluate NeuralPOM in a closed, wind-driven double-gyre ocean basin spanning 30°N to 50°N across a 2500-km wet interior with a flat 2000-m bottom and 15 active terrain-following σ-layers (16 vertical storage levels; Fig. 7 and Appendix D). Steady sinusoidal zonal wind stress forces a subtropical and subpolar gyre pair separated by an unstable eastward jet and an energetic mesoscale eddy field. Potential temperature is initialized from an exponential vertical stratification profile and salinity from a uniform value of 35 in model units with zero surface heat or freshwater fluxes. Because salinity experiences no external forcing or boundary restoring, variations in the fine-grid reference salinity field (standard deviation ~ 0.08) arise purely from finite-volume advection and vertical turbulent mixing along σ-surfaces; any unphysical tracer source or vertical mixing perturbation introduced during a forecast therefore produces a distinct signature in salinity.

For each dynamical framework (Explicit and Implicit), we integrated an eight-member fine-grid reference ensemble on a $354\times322\times16$ grid (~ 7 km horizontal spacing) for 9000 model days per member initialized from distinct divergence-free velocity perturbations, exceeding 100,000 model days across both ensembles. Volume-weighted conservative restriction onto the $90 \times 82 \times 16$ coarse forecast grid (~ 28 km spacing, where baroclinic eddies are only marginally resolved) archives complete daily 45-variable state checkpoints alongside 20-minute subdaily targets during the first four hours of each day (Appendix B.1). Ensemble members 0–6 over days 1000–7000 provide training trajectories, member 7 over days 7000–8000 provides validation data, and all out-of-sample evaluations are initialized from member 7 at 16 evenly spaced forecast origins $d_r = 8000 + 32r$ ($r = 0, \ldots, 15$) and integrated for 100 days.

3.2 Baselines and evaluation protocol

Our main experimental design crosses the two dynamical frameworks (Implicit and Explicit), two coupling intervals ($\Delta = 1$ day and $\Delta = 1$ h), and three training stages (Pretrain, Midtrain, and Posttrain; optimizer schedules and update budgets are summarized in Appendix B.5, Table 6). Coarse-grid physical and hybrid integrations run at $\Delta t = 600$ s in 32-bit floating point, while volume-weighted root-mean-square errors (RMSE) and radially averaged horizontal velocity wavenumber spectra are accumulated in 64-bit precision (Appendix D.1).

On the uncorrected $90 \times 82 \times 16$ POM solver, we compare three horizontal subgrid dissipation closures (Appendix D.2). The primary uncorrected physical baseline in each dynamical framework uses standard second-order harmonic *Smagorinsky* viscosity and diffusivity ($c_m = 0.05, c_c = 0.01$). We additionally evaluate scale-selective fourth-order *Biharmonic* momentum dissipation ($\gamma = 2$) and a kinetic-energy *Backscatter* closure that couples biharmonic dissipation to a column subgrid energy reservoir recycling a fraction $\beta = 0.5$ of dissipated kinetic energy into the resolved flow via anti-viscous harmonic forcing. Alongside these physical closures, we train five direct spatiotemporal neural architectures—*Triton* [29], a *ResUNet* sharing the residual U-Net architecture of the hybrid corrector, *SimVP* [30], *ConvLSTM* [31], and *PastNet* [32]—that step the 313-channel daily state directly from day d to day $d + 1$ without a numerical solver. Each direct neural baseline is trained through both single-step pretraining (*Pre*) and 10-day closed-loop rollout training (*Post*) and evaluated over 100-day forecasts from the same 16 test origins against the semi-implicit reference (Appendix E).

All NeuralPOM networks are trained with $c = 1$ in Eq. (2.4) and evaluated at inference with frozen network weights under three tracer update rules: the unadjusted zero-budget update (*Undamped*, $c = 1$), *Density-Normal Damping* (*DND*, $c = 3/4$), and an isotropic control (*Uniform*) that scales the entire constrained tracer increment z by a fixed scalar factor $\kappa \in [0.904, 0.931]$ calibrated on development states to match the mean volume-weighted standardized tracer increment norm of DND in each configuration (Appendix C.2). To separate backpropagating gradients through the differentiable dynamical framework during Midtrain from multi-step rollout exposure and closed-loop Posttrain, we compare three training routes across all 16 test origins under Undamped and DND rules (Appendix B.5): the full three-stage procedure (*Full*, Pre $\rightarrow$ Mid $\rightarrow$ Post), skipping Midtrain (*Skip*, Pre $\rightarrow$ Post), and a detached-gradient control (*Detached*, Pre $\rightarrow \text{Mid}_{\text{detached}} \rightarrow$ Post) that runs the identical 4-step Midtrain trajectories and update budgets while detaching the state before each physical interval $\mathcal{P}^{\Delta}$. Finally, to diagnose how a single tracer update propagates through the primitive equations without subsequent neural intervention, we sample 36 pre-correction states (lead days 1, 10, and 30 from origins 8000, 8160, and 8320 across all four Posttrain configurations). At $t = 0$, we apply the identical baroclinic velocity update, branch the single tracer update into *no-TS* (omitting the tracer increment), *Undamped*, *Uniform*, and *DND*, and integrate the uncorrected physical framework forward for 72 hours.

4 Forecast Results

4.1 Whole-state forecast skill across dynamical frameworks

Across 100-day forecasts initialized from the 16 out-of-sample test origins, NeuralPOM improves whole-state forecast skill relative to the uncorrected coarse-grid Smagorinsky baseline in both the semi-implicit (Implicit) and explicit mode-split (Explicit) dynamical frameworks (Table 1 and Fig. 2). Under daily coupling ($\Delta = 1$ day), both DND and Undamped deployments lower 100-day root-mean-square error

Table 1. Whole-state 100-day forecast skill relative to the coarse-grid Smagorinsky baseline. Percentage change in pooled 100-day RMSE across the 16 test origins relative to the uncorrected Smagorinsky physical model of the same dynamical framework (Implicit or Explicit); negative values indicate error reductions. Hybrid rows report the final Posttrain checkpoints deployed with Density-Normal Damping (DND, $c = 3/4$) and the undamped tracer update (Undamped, $c = 1$). Seawater density ρ is evaluated diagnostically via the nonlinear POM equation of state.

Framework	Configuration	Interval	u	v	SSH	T	S	ρ (EOS)
Implicit	Biharmonic (physics)	–	+10.94	+8.39	+7.60	+6.84	+15.70	+7.37
Implicit	Backscatter (physics)	–	+11.44	+8.82	+7.27	+6.76	+16.53	+7.33
Implicit	NeuralPOM (Undamped)	1 day	−7.03	−5.27	−6.72	−9.90	−14.72	−7.75
Implicit	NeuralPOM (DND)	1 day	−6.45	−4.97	−6.21	−9.59	−16.56	−7.87
Implicit	NeuralPOM (Undamped)	1 h	−2.01	−0.53	−3.53	+3.16	+58.66	+3.63
Implicit	NeuralPOM (DND)	1 h	−9.47	−7.32	−12.12	−9.04	−0.93	−7.11
Explicit	Biharmonic (physics)	–	+7.54	+6.29	+5.05	+3.67	+4.60	+3.70
Explicit	Backscatter (physics)	–	+8.62	+7.20	+5.69	+4.17	+5.19	+4.20
Explicit	NeuralPOM (Undamped)	1 day	−9.62	−4.92	−9.03	−11.83	−16.17	−9.85
Explicit	NeuralPOM (DND)	1 day	−9.46	−5.13	−8.96	−11.98	−17.95	−10.56
Explicit	NeuralPOM (Undamped)	1 h	−5.69	−3.33	−7.89	+5.38	+90.57	+3.46
Explicit	NeuralPOM (DND)	1 h	−10.35	−8.64	−13.49	−8.15	+7.98	−5.63

(RMSE) across all five prognostic variables, reducing horizontal velocity and sea-surface height (SSH) errors by 5%–10%, potential temperature error by 10%–12%, salinity error by 15%–18%, and diagnostic equation-of-state (EOS) density error by 8%–11% (Fig. 2b). Even without direct neural updates to η, SSH errors decrease by 6%–9% because updates to baroclinic velocity shear and interior buoyancy stratification adjust the sea surface through barotropic–baroclinic mode coupling during the subsequent physical integration.

Comparing daily and hourly coupling illustrates the trade-off between frequent circulation guidance and baroclinic sensitivity to tracer updates. When deployed with DND, hourly coupling ($\Delta = 1$ h) produces the largest circulation and sea-level error reductions in both frameworks, lowering 100-day RMSE by 7%–10% in horizontal velocity and 12%–13% in SSH alongside 8%–9% reductions in temperature error and 6%–7% reductions in density error. Without directional damping (Undamped), however, hourly tracer updates degrade the unforced salinity field by 59%–91% and increase temperature and density errors by 3%–5% relative to Smagorinsky; as shown in Section 5, DND prevents this degradation by attenuating the density-modifying component of the tracer increment. On the uncorrected coarse grid, replacing harmonic Smagorinsky diffusion with scale-selective Biharmonic viscosity or kinetic-energy Backscatter increases 100-day RMSE across all variables by 4%–17% in both frameworks (Table 1), because energizing marginally resolved eddies without data-driven phase corrections increases double-penalty errors against the reference trajectory.

4.2 Comparison with direct neural forecasters

We next compare daily NeuralPOM (DND) against the Pretrain (Pre) and Posttrain (Post) checkpoints of the five direct 313-channel neural architectures on the semi-implicit reference (Table 2; lead-window splits are reported in Appendix E). Across the 100-day evaluation window, daily NeuralPOM achieves the lowest RMSE in every prognostic field, improving upon the strongest direct neural baseline in each variable by 5%–12% in horizontal velocity and SSH, 28% in temperature, and 74% in salinity. With the sole exception of meridional velocity in SimVP Post, every direct neural forecast in Table 2 incurs a larger 100-day RMSE than the uncorrected coarse-grid Smagorinsky solver. Moreover, because direct neural

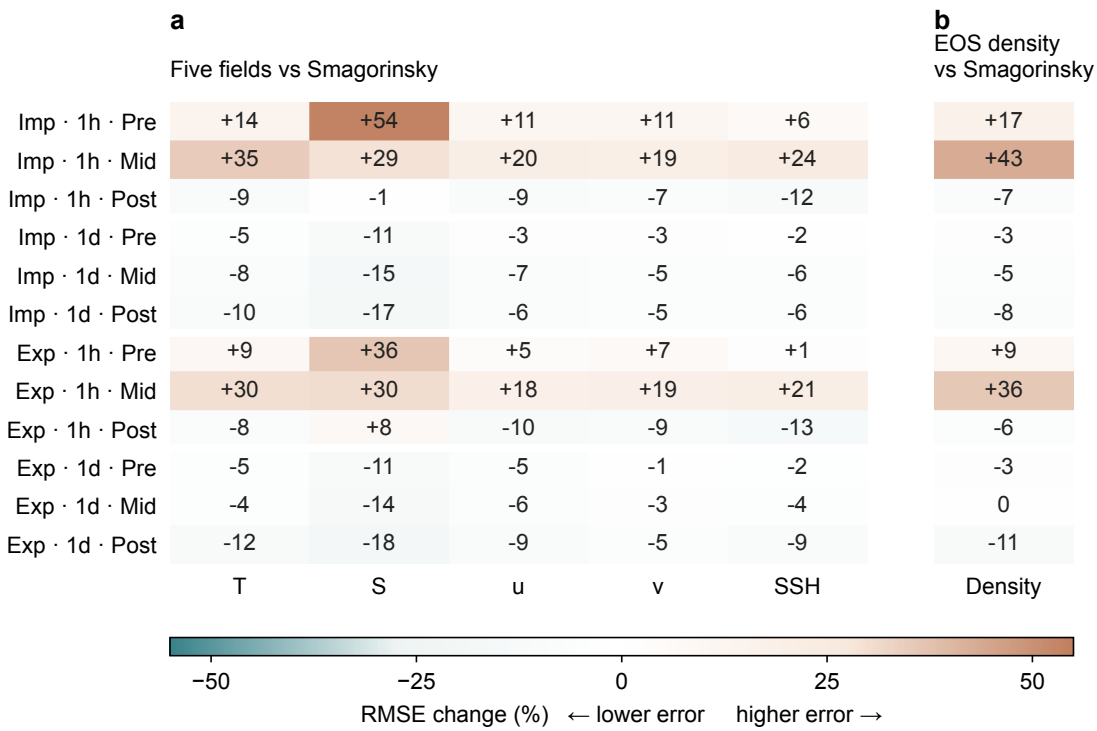


Figure 2. Forecast skill across training stages and correction intervals. a, Percentage change in 100-day RMSE relative to the Smagorinsky physical baseline for all twelve hybrid configurations deployed with DND. **b,** Corresponding percentage change in 100-day EOS density RMSE. Negative values indicate improvements over Smagorinsky; annotations are rounded to whole percentages. Imp and Exp denote the semi-implicit and explicit frameworks.

models do not enforce finite-volume layer conservation, their 100-day salinity RMSE exceeds that of the Smagorinsky baseline by a factor of three to eight.

Radially averaged horizontal velocity spectra over forecast days 51–100 clarify the physical differences between hybrid and purely data-driven rollouts (Table 3). Direct neural models trained only with single-step pretraining either lose most of their kinetic energy over 100-day integrations or develop unaligned small-scale energy growth (Appendix E). Incorporating 10-day closed-loop rollout training (Post) restores bulk kinetic energy across the five direct architectures ($R \approx 0.56$–0.79), but damps near-grid mesoscale variability ($0.25 < k \leq 0.5$ cpg) to $R_{\text{near-grid}} \approx 0.07$–$0.29$, compared with 1.00 in the Smagorinsky solver and 1.13 in hourly NeuralPOM (DND). Consequently, all direct neural architectures exhibit higher full-spectrum velocity error ($Q \approx 1.20$–1.42) over days 51–100 than both the uncorrected physical model ($Q = 1.18$) and NeuralPOM ($Q \approx 1.01$–1.11).

4.3 Eddy variability, phase alignment and spatial error structure

To separate amplitude errors from eddy phase errors across wavenumbers, we define the predicted-to-reference kinetic-energy ratio $R = E^p/E^r$, cross-spectral phase alignment $A = C/\sqrt{E^p E^r}$, and normalized error energy $Q = E^e/E^r$ (Appendix D.4). These quantities satisfy the spectral identity

$$Q = R + 1 - 2A\sqrt{R}, \tag{4.1}$$

which decomposes any change in spectral error energy relative to the Smagorinsky baseline additively into an amplitude contribution $\Delta Q_{\text{amp}} = \Delta R$ and a phase-alignment contribution $\Delta Q_{\text{align}} = -2\Delta(A\sqrt{R})$.

Table 2. Comparison of hybrid NeuralPOM and direct neural forecasters. Pooled 100-day RMSE on the semi-implicit reference across the 16 test origins. Shading and boldface mark the best value in each column and underlining marks the second best, ranked on unrounded scores. Pre and Post denote the frozen Pretrain and Posttrain checkpoints of each direct 313-channel architecture; salinity is in model units.

Configuration	$u \downarrow$ $\mathrm{m\,s^{-1}}$	$v \downarrow$ $\mathrm{m\,s^{-1}}$	SSH $\downarrow$ m	$T \downarrow$ °C	$S \downarrow$
`Smagorinsky`	<u>0.07926</u>	0.08197	<u>0.11970</u>	<u>0.29351</u>	<u>0.03382</u>
Triton Pre	0.09935	0.09526	0.21544	0.79294	0.21825
Triton Post	0.08886	0.08663	0.14570	0.36668	0.10914
ResUNet Pre	0.10189	0.09767	0.16460	0.45770	0.12531
ResUNet Post	0.08267	0.08401	0.12812	0.60935	0.22609
SimVP Pre	0.09434	0.09123	0.22382	0.79226	0.21567
SimVP Post	0.08452	<u>0.08180</u>	0.13804	0.63266	0.25703
ConvLSTM Pre	0.10088	0.10351	0.29508	0.96630	0.22385
ConvLSTM Post	0.10222	0.09665	0.18062	0.51404	0.10723
PastNet Pre	0.14249	0.12674	0.23017	0.61166	0.17605
PastNet Post	0.10278	0.10150	0.17001	0.77017	0.26912
NeuralPOM Post-`DND` (ours)	**0.07415**	**0.07789**	**0.11227**	**0.26537**	**0.02822**

Table 3. Layer-weighted velocity spectral statistics over forecast days 51–100. Full-scale energy ratio R, phase alignment A, normalized error energy $Q = R + 1 - 2A\sqrt{R}$, additive decomposition of $\Delta Q = Q - Q_{\mathrm{Smag}}$ into amplitude (ΔR) and alignment ($-2\Delta(A\sqrt{R})$) contributions relative to Smagorinsky of the same dynamical framework, and near-grid energy ratio $R_{\text{near-grid}}$ ($0.25 < k \le 0.5$ cpg). Hybrid rows show the final Posttrain models deployed with DND; direct AI models are evaluated on the semi-implicit reference.

Model	Interval	R	A	$Q \downarrow$	ΔQ_{amp}	$\Delta Q_{\mathrm{align}}$	$R_{\text{near-grid}}$
Smagorinsky (Implicit)	–	0.559	0.253	1.181	–	–	0.998
Biharmonic (Implicit)	–	0.819	0.229	1.404	+0.260	−0.037	2.231
Backscatter (Implicit)	–	0.840	0.237	1.405	+0.281	−0.057	2.512
NeuralPOM Implicit (DND)	1 day	0.595	0.315	1.109	+0.036	−0.108	0.378
NeuralPOM Implicit (DND)	1 h	0.604	0.385	1.006	+0.045	−0.220	1.134
Smagorinsky (Explicit)	–	0.528	0.222	1.205	–	–	0.836
Biharmonic (Explicit)	–	0.753	0.205	1.397	+0.226	−0.033	1.700
Backscatter (Explicit)	–	0.776	0.197	1.428	+0.248	−0.025	1.884
NeuralPOM Explicit (DND)	1 day	0.581	0.287	1.143	+0.053	−0.115	0.365
NeuralPOM Explicit (DND)	1 h	0.504	0.340	1.022	−0.024	−0.159	1.033
Triton Post (direct AI)	1 day	0.562	0.241	1.201	+0.003	+0.017	0.072
SimVP Post (direct AI)	1 day	0.789	0.329	1.204	+0.230	−0.207	0.090
ResUNet Post (direct AI)	1 day	0.753	0.306	1.222	+0.194	−0.153	0.121
ConvLSTM Post (direct AI)	1 day	0.600	0.160	1.352	+0.041	+0.129	0.191
PastNet Post (direct AI)	1 day	0.741	0.185	1.421	+0.181	+0.059	0.294

Table 4. Staged training progression and controlled route ablations under DND deployment. For each configuration deployed with DND ($c = 3/4$), Mid/Pre and Post/Mid report percentage changes in 100-day RMSE between successive stages (negative values indicate error reductions). Full vs. Skip and Full vs. Detached report relative 100-day RMSE reductions $100\% \times (1 - E_{\text{Full}}/E_{\text{control}})$ of the full route (Pre → Mid → Post) over skipping Midtrain (Pre → Post) and detached-gradient Midtrain (Pre → $\text{Mid}_{\text{detached}}$ → Post) at matched update budgets (positive values favor Full).

Framework	Interval	Comparison	u	v	SSH	T	S
Implicit	1 day	Mid/Pre (Δ%)	−3.47	−2.33	−4.62	−3.28	−4.66
		Post/Mid (Δ%)	+0.19	+0.15	−0.04	−2.07	−1.49
		Full vs. Skip (% gain)	+0.09	+0.72	+0.39	+0.21	+0.34
		Full vs. Detached (% gain)	−0.38	+0.06	−0.20	−0.13	−0.10
Explicit	1 day	Mid/Pre (Δ%)	−0.56	−2.28	−1.66	+0.63	−3.38
		Post/Mid (Δ%)	−4.12	−1.90	−5.50	−8.39	−4.46
		Full vs. Skip (% gain)	+0.29	+0.10	+0.33	+0.33	+0.87
		Full vs. Detached (% gain)	−0.74	−0.54	−1.05	−0.85	−0.40
Implicit	1 h	Mid/Pre (Δ%)	+8.39	+7.74	+17.02	+18.70	−15.98
		Post/Mid (Δ%)	−24.55	−22.23	−29.18	−32.63	−23.28
		Full vs. Skip (% gain)	−1.58	−1.59	−3.71	+1.02	+3.58
		Full vs. Detached (% gain)	+0.06	+0.39	−1.82	+0.96	+9.71
Explicit	1 h	Mid/Pre (Δ%)	+12.35	+10.60	+18.87	+19.36	−4.35
		Post/Mid (Δ%)	−24.00	−23.02	−28.21	−29.17	−17.22
		Full vs. Skip (% gain)	+3.23	+2.67	+3.46	+8.42	+41.15
		Full vs. Detached (% gain)	+4.66	+4.88	+5.09	+8.80	+38.75

Across all four hybrid configurations, NeuralPOM (DND) maintains the bulk eddy kinetic energy of the coarse-grid physical framework ($R \approx 0.50$–0.60 versus 0.53–0.56 for Smagorinsky) while raising full-spectrum phase alignment A from 0.22–0.25 to 0.29–0.39, lowering late-window Q to 1.01–1.14 (Table 3 and Fig. 3). The additive decomposition in Table 3 shows that the reduction in Q arises from improved mesoscale phase alignment ($\Delta Q_{\text{align}} = -0.11$ to -0.22). Equation (4.1) also explains why direct neural models can achieve low near-grid MSE while losing small-scale physical structure: when phase alignment vanishes at high wavenumbers ($A \simeq 0$), $Q \simeq 1 + R$, so damping near-grid variance ($R \to 0$) drives near-grid Q toward unity [9, 10]. Over the full wavenumber spectrum, where mesoscale energy dominates, both daily and hourly NeuralPOM achieve lower spectral error Q than all physical closures and direct neural models, and hourly coupling preserves near-grid kinetic energy ($R_{\text{near-grid}} \approx 1.03$–$1.13$). Finally, decomposing the three-dimensional tracer MSE into domain-mean, vertical profile, and horizontal frontal components (Eq. (D.3); Fig. 6 and Appendix D.3) shows that horizontal eddy and frontal structure accounts for over 99.6% of the 100-day temperature MSE; in the unforced salinity field, horizontal structure errors account for 76%–79% of total MSE, accompanied by vertical profile drift (11%–18%) and domain-mean drift (6%–11%).

4.4 Contributions of staged training and route ablation

Comparing checkpoints across the three training stages (Pretrain → Midtrain → Posttrain) and the three controlled training routes (*Full*, *Skip*, and *Detached*) at matched update budgets shows how multi-step differentiable training and closed-loop rollout training interact with the coupling interval and free-surface solver (Table 4; Appendix B.5).

The progression across training stages reflects the timescale of the Midtrain window relative to the local inertial period $2\pi/f \approx 19\,\text{h}$. Under daily coupling, each step exceeds the inertial period, and 4-day

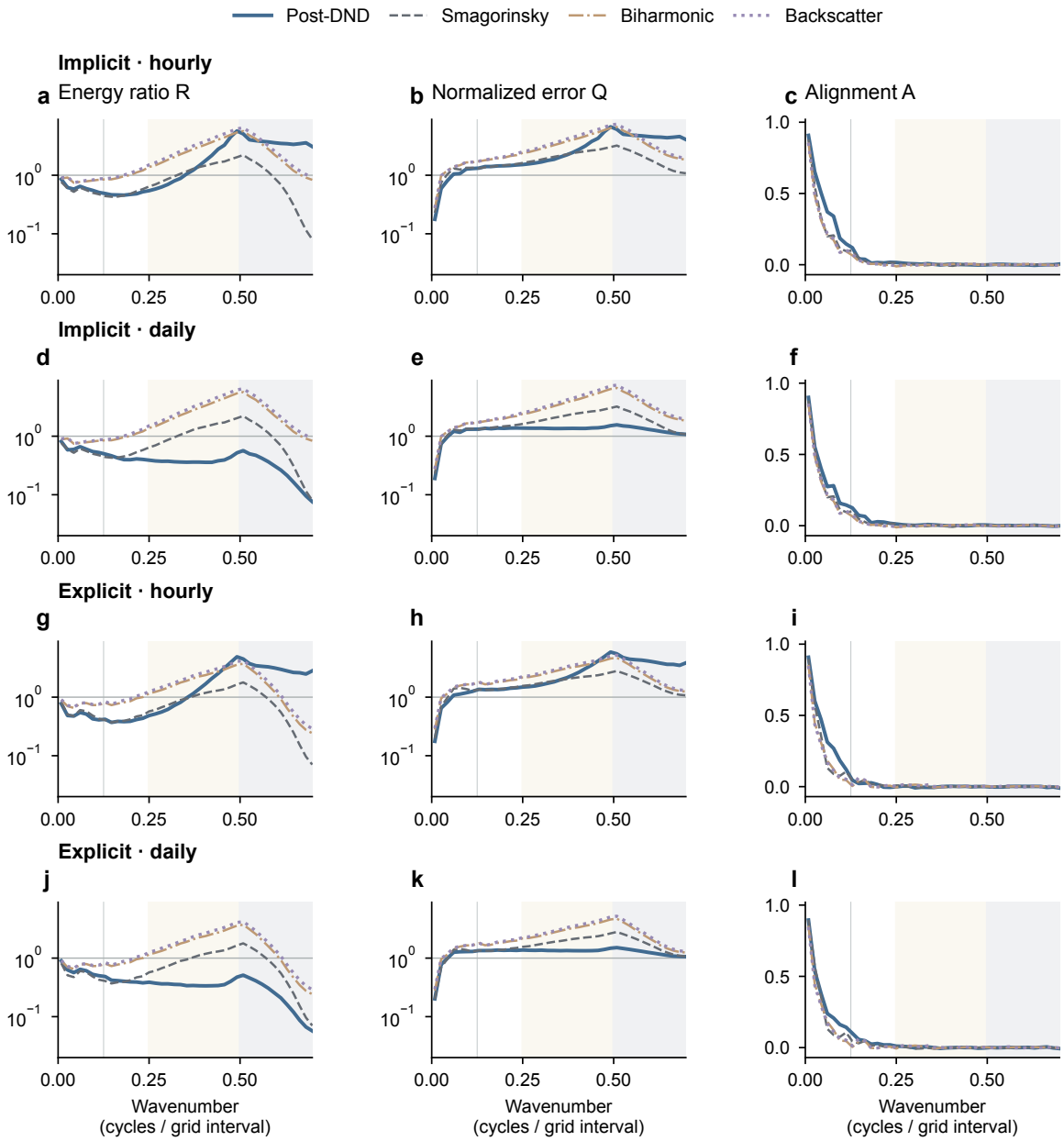


Figure 3. Wavenumber velocity spectra of the late forecast. Rows correspond to the four Posttrain hybrid configurations deployed with DND over days 51–100. Columns display the predicted-to-reference kinetic-energy ratio $R(k)$, the error-to-reference energy ratio $Q(k)$ and the phase alignment $A(k)$ as functions of radial wavenumber k (cycles per grid interval), weighted by layer thickness. Shaded bands denote the near-grid scale ($0.25 < k \leq 0.5$ cpg) and corner wavenumbers ($k > 0.5$ cpg).

differentiable Midtrain followed by 10-day closed-loop Posttrain progressively reduces 100-day RMSE across both frameworks, yielding cumulative Post/Pre reductions of 2%–8% under DND. Under hourly coupling, the 4-hour Midtrain window spans less than a quarter of an inertial period: optimizing across four hourly steps reduces 100-day salinity RMSE by 4%–16% but increases mesoscale kinetic energy (R rising from 0.73 to 1.08–1.09 over days 51–100), temporarily elevating velocity, SSH, and temperature RMSE. Subsequent 10-day closed-loop Posttrain over 240 hourly cycles per rollout damps this excess kinetic energy back to $R \approx 0.50$–0.60, raises phase alignment A to 0.34–0.39, and reduces 100-day RMSE relative to Midtrain by 17%–33% across all five prognostic variables.

The three-route ablation (*Full*, *Skip*, and *Detached*) separates multi-step rollout exposure from backpropagating gradients through the differentiable physical framework during Midtrain (Table 4). Under daily coupling, where each 24-hour step spans a full inertial adjustment cycle, 10-day closed-loop Posttrain dominates long-range skill and all three routes agree within 1.1% in RMSE and 1.0% in spectral error Q. Under hourly coupling, backpropagating gradients across physical steps is decisive. In the explicit mode-split framework, where 30 barotropic substeps per baroclinic step propagate fast external gravity waves, 4-step rollouts with detached gradients (*Detached*) improve salinity by only 3.9% over *Skip* while slightly degrading velocity, SSH, and temperature (−0.4% to − 2.3%). Retaining cross-physics gradients (*Full* versus *Detached*) reduces 100-day RMSE across all five fields—by roughly 5% in velocity and SSH, 9% in temperature (12% under Undamped), and 39% in salinity (32% under Undamped)—accounting for 88%–89% of the salinity MSE reduction over *Skip* and lowering full-spectrum velocity error Q by 10% (and near-grid Q by 14%–19%). In the semi-implicit hourly framework, where implicit free-surface damping filters external gravity waves at each 600-s step, detached rollouts degrade salinity by 6.8% relative to *Skip*, whereas retaining cross-physics gradients (*Full* versus *Detached*) reduces 100-day salinity RMSE by 10%–12% and temperature RMSE by 1%–2%.

5 Baroclinic Feedback of Tracer Corrections and Density-Normal Damping

5.1 Baroclinic pressure-gradient feedback and density-normal decomposition

As shown in Table 1, applying undamped tracer updates ($c = 1$) at daily intervals improves all five prognostic variables over 100-day forecasts, whereas applying undamped updates at hourly intervals degrades temperature by 3%–5% and salinity by 59%–91% relative to the coarse-grid Smagorinsky baseline. Wavenumber spectra over forecast days 51–100 show that hourly Undamped integrations accumulate excess kinetic energy at near-grid scales ($R_{\text{near-grid}} = 1.83$–$2.26$) and corner wavenumbers ($k > 0.5$ cpg, $R_{\text{corner}} = 5.85$–9.51), whereas daily integrations maintain $R_{\text{near-grid}} \leq 0.43$. In the three-dimensional primitive equations, potential temperature and salinity enter the momentum and turbulence equations through the diagnostic density $\rho(T, S, P)$ evaluated from the nonlinear equation of state (EOS). When temperature and salinity increments contain uncoordinated small-scale residuals $(\delta T, \delta S)$, they project onto in-situ density anomalies $\delta\rho$. Under hourly cycling, horizontal gradients of $\delta\rho$ repeatedly introduce ageostrophic hydrostatic baroclinic pressure-gradient tendencies at near-grid scales before geostrophic adjustment can radiate or dissipate them, while vertical gradients $-\partial_\sigma \delta\rho$ perturb the buoyancy frequency N^2 and Mellor–Yamada vertical diffusivities (K_H, K_M), distorting vertical tracer profiles.

To suppress this baroclinic feedback while preserving layerwise heat and salt conservation, we decompose the standardized layer-mean-preserving tracer increment $z_i = (\delta T_i/s_T, \delta S_i/s_S)^\top \in \mathbb{R}^2$ in wet cell i relative to the local EOS density gradient. On each active σ-layer, let $w_i > 0$ $(\sum_i w_i = 1)$ denote the normalized cell-volume weight on the zero-mean space $\mathcal{H}_0 = \{z : \sum_i w_i z_i = 0\}$ equipped with the weighted inner

product $\langle u, v\rangle_w = \sum_i w_i u_i^\top v_i$. Letting $g_i = (s_T \partial_T \rho_i, s_S \partial_S \rho_i)^\top$ denote the scaled EOS density gradient at the pre-correction state and $n_i = g_i / \|g_i\|_2$ the local unit density-normal vector, the layer-mean-preserving density-modifying subspace on each active σ-layer is

$$\mathcal{S} = \left\{ p : \; p_i = \beta_i n_i, \; \sum_i w_i n_i \beta_i = 0 \right\}, \tag{5.1}$$

which contains all pointwise density-normal tracer increments that conserve layer-integrated heat and salt (the closed-form w-orthogonal projection $E : \mathcal{H}_0 \to \mathcal{S}$ is derived in Appendix C.1). This projection separates any conserved tracer increment $z \in \mathcal{H}_0$ into a density-modifying (diapycnal) component Ez and an orthogonal density-compensated (isopycnal) component $(I - E)z$. To first order in the EOS, the density-compensated component leaves horizontal and vertical density gradients unchanged and decays under advection and diffusion, whereas the density-modifying component alters buoyancy stratification and generates hydrostatic baroclinic pressure-gradient tendencies (Appendix C).

Density-Normal Damping (DND) scales the density-modifying component by $c \in [0, 1]$ while retaining the orthogonal density-compensated component in full:

$$z_c = (I - E)z + cEz. \tag{5.2}$$

With $c = 3/4$, DND reduces the squared norm of the density-modifying projection by $1 - (3/4)^2 = 43.75\%$ while preserving 100% of the density-compensated water-mass correction and maintaining exact layerwise heat and salt conservation (Appendix C).

5.2 Single-perturbation propagation over 72 hours

We test this physical mechanism directly using the 36-state, four-branch single-perturbation integration experiment across all four frozen Posttrain configurations (Fig. 4), comparing no-TS, Undamped ($c = 1$), norm-matched Uniform ($\kappa \in [0.904, 0.931]$), and DND ($c = 3/4$) over 72 hours of free physical integration following a single state update at $t = 0$. At $t = 0$, evaluating the model's discrete hydrostatic pressure-gradient operator (Eqs. (A.10)–(A.11)) on the updated density field shows that DND reduces the instantaneous baroclinic pressure-gradient tendency MSE relative to Undamped by 43.7%–46.2% across all four configurations—closely matching the linearized 43.75% scaling—whereas norm-matched Uniform scaling reduces it by only 13.3%–18.3% (Fig. 4a). Over the ensuing 72 hours of uncorrected physical integration, this initial reduction translates into a 43%–46% decrease in tracer-induced horizontal velocity perturbation energy across all configurations and lead times.

Tracking the horizontal velocity RMSE across the 72-hour physical integration explains why DND produces large gains under hourly coupling while leaving daily circulation skill nearly unchanged (Fig. 4b,c). In both hourly models, a single Undamped tracer update excites ageostrophic velocity perturbations that increase horizontal velocity RMSE over 1–72 hours relative to omitting the tracer update (no-TS; Fig. 4b), and DND consistently lowers velocity RMSE relative to both Undamped and norm-matched Uniform across all lead times (Fig. 4c). In both daily models, by contrast, geostrophic adjustment over 24–72 hours allows the tracer-induced baroclinic response to reduce background velocity error relative to no-TS (Fig. 4b), so daily tracer updates improve the resolved circulation without requiring strong diapycnal damping.

5.3 Closed-loop 100-day verification against uniform scaling

Evaluating Undamped, norm-matched Uniform, and DND across closed-loop 100-day forecasts from all 16 test origins with identical frozen Posttrain network weights demonstrates that selectively damping

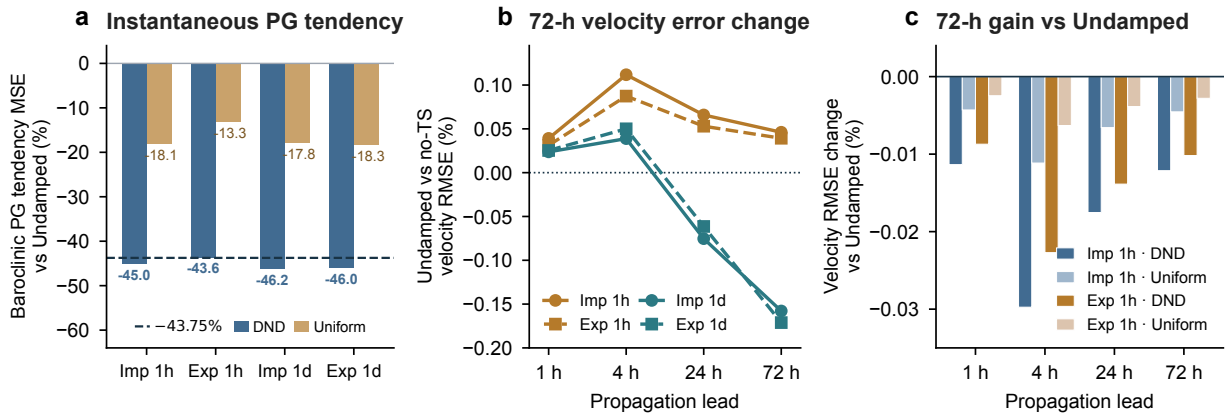


Figure 4. Single-perturbation physical response and 72-hour propagation across 36 forecast states. a, Instantaneous percentage change in EOS-induced baroclinic pressure-gradient tendency MSE relative to Undamped at $t = 0$ across all four Posttrain configurations; the dashed line marks the linearized reduction $-(1 - (3/4)^2) = -43.75\%$. **b,** Percentage change in horizontal velocity RMSE of a single Undamped tracer update relative to omitting the tracer update (no-TS) over 72 hours of uncorrected physical integration. **c,** Percentage change in horizontal velocity RMSE of DND and norm-matched Uniform relative to Undamped across the two hourly configurations over 1 to 72 hours.

the density-modifying component governs long-range hybrid stability under frequent coupling (Table 5 and Fig. 5). In both hourly configurations, DND reduces 100-day RMSE relative to Undamped by 5%–9% in horizontal velocity and SSH, 12%–13% in temperature, 38%–43% in salinity, and 9%–10% in EOS density—roughly four to eight times the reductions achieved by norm-matched Uniform scaling (1%–6%; Table 5 and Fig. 5a). Because all three deployments share the same network weights, baroclinic velocity updates, and zero direct free-surface modifications, the improvements in velocity and SSH under DND result entirely from suppressing spurious baroclinic pressure-gradient tendencies and vertical mixing perturbations during closed-loop cycling.

Late-window velocity spectra (days 51–100) and three-dimensional tracer error decompositions corroborate both physical pathways (Fig. 5b). In the hourly configurations, replacing Undamped or Uniform updates with DND restores the near-grid velocity energy ratio $R_{\text{near-grid}}$ from 1.83–2.26 and 1.64–2.02 down to 1.03–1.13, increases full-spectrum phase alignment A from 0.29 to 0.34–0.39, and lowers spectral error energy Q from 1.13–1.17 to 1.01–1.02. At the same time, nearly half (42%–51%) of the 100-day hourly salinity MSE reduction between Undamped and DND comes from eliminating vertical profile drift caused by buoyancy-driven perturbations to the turbulence closure, with the remainder coming from improved horizontal frontal structure. In the daily configurations, where tracer updates undergo 24 hours of geostrophic adjustment between corrections, DND maintains 100-day velocity, SSH, and temperature RMSE within 0.6% of Undamped while reducing salinity RMSE by 2.1%.

6 Discussion

NeuralPOM shows that a three-dimensional, terrain-following primitive-equation ocean circulation model and a neural state corrector can be coupled and trained to outperform both coarse-grid subgrid closures and direct autoregressive neural forecasters over 100-day integrations. On an eddy-permitting coarse grid, conventional subgrid parameterizations cannot recover the spatial phase of unresolved mesoscale eddies

Table 5. Closed-loop 100-day comparison of directional damping versus uniform scaling on frozen Posttrain networks. Percentage change in 100-day RMSE relative to Undamped ($c = 1$) across all 16 evaluation origins for identical network weights, initial states, and reference trajectories. DND retains $c = 3/4$ of the layer-mean-preserving density-modifying (diapycnal) tracer component while preserving 100% of the orthogonal density-compensated (isopycnal) component; Uniform scales the entire tracer increment by a fixed factor $\kappa \in [0.904, 0.931]$ calibrated to match the mean volume-weighted standardized norm of DND.

Framework	Interval	Rule	u	v	SSH	T	S	ρ
Implicit	1 h	Uniform	−2.42	−2.24	−2.38	−3.10	−5.87	−3.24
		DND	−7.62	−6.82	−8.90	−11.83	−37.56	−10.37
Explicit	1 h	Uniform	−1.10	−1.22	−1.16	−2.48	−5.23	−2.22
		DND	−4.94	−5.50	−6.08	−12.83	−43.34	−8.78
Implicit	1 day	Uniform	+0.18	+0.04	+0.18	−0.02	−0.18	−0.16
		DND	+0.63	+0.32	+0.55	+0.35	−2.16	−0.13
Explicit	1 day	Uniform	+0.21	−0.15	+0.18	−0.07	−0.14	−0.24
		DND	+0.18	−0.22	+0.08	−0.17	−2.13	−0.79

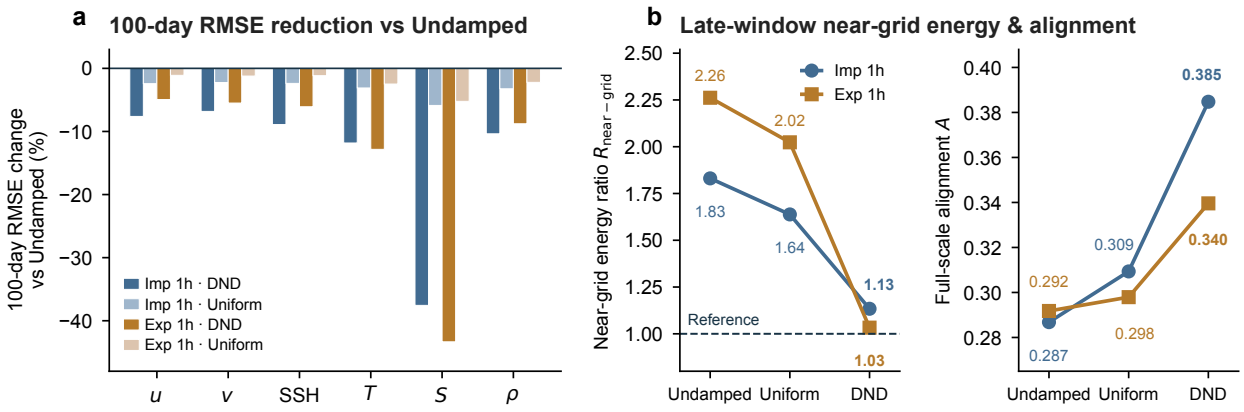


Figure 5. Closed-loop 100-day verification of DND versus norm-matched Uniform in hourly coupling. a, Percentage change in 100-day RMSE across all five primary variables and in-situ EOS density (ρ) for DND and norm-matched Uniform relative to Undamped across all 16 evaluation origins on frozen Posttrain networks. **b,** Late-window (days 51–100) depth-weighted near-grid velocity energy ratio $R_{\text{near-grid}}$ (left; dashed line marks the fine-grid reference $R = 1$) and full-scale phase alignment A (right) across Undamped, Uniform, and DND.

from resolved strain rates alone, whereas unconstrained direct neural forecasters damp small-scale variability and accumulate volume and tracer conservation errors during extended rollouts. By retaining explicit mode-split and semi-implicit primitive-equation frameworks to enforce volume continuity, hydrostatic and geostrophic balance, and vertical turbulent mixing while applying constrained baroclinic velocity and layerwise tracer updates at regular intervals, NeuralPOM reduces forecast errors across horizontal velocity, sea-surface height, potential temperature, salinity, and in-situ density while maintaining realistic mesoscale kinetic energy. Stable long-range performance requires both the conservation-constrained state update—comprising column-mean shear projection, layer-integrated heat and salt conservation, leapfrog-consistent updates across current and backward time levels, and immediate equation-of-state density evaluation—and the three-stage training procedure. In particular, controlled route ablations show that backpropagating gradients through the differentiable dynamical framework during 4-step Midtrain—rather than multi-step rollout exposure with detached gradients—drives the long-range thermohaline and circulation gains under hourly explicit mode-split coupling.

The diagnostic perturbation and closed-loop experiments clarify how thermohaline state updates interact with primitive-equation dynamics under frequent coupling. Because potential temperature and salinity force horizontal momentum and vertical turbulence through the nonlinear equation of state $\rho(T, S, P)$, decomposing each layerwise tracer increment relative to the local EOS density gradient separates a density-compensated (isopycnal) component, which leaves buoyancy stratification invariant to first order and decays under advection and diffusion, from a layer-mean-preserving density-modifying (diapycnal) component, which alters hydrostatic baroclinic pressure gradients and vertical buoyancy stratification. When corrections are applied hourly—much faster than the $\sim$ 19 h local inertial period required for geostrophic adjustment—uncoordinated residuals in temperature and salinity project onto the density-modifying component, exciting near-grid ageostrophic velocity variance and perturbing Mellor–Yamada vertical mixing. Density-Normal Damping (DND) mitigates this feedback by attenuating the squared norm of the density-modifying component by 43.75% while preserving the density-compensated water-mass correction in full, yielding four- to eightfold larger 100-day error reductions than norm-matched isotropic scaling in hourly forecasts. Physically, DND serves a role analogous to incremental analysis updates [33] and tangent-linear balance constraints in operational ocean data assimilation by limiting imbalanced baroclinic forcing during frequent state updates.

Several extensions follow naturally from these results. First, because the coarse grid resolves roughly 55%–60% of the fine-grid eddy kinetic energy and deterministic MSE objectives penalize phase-misaligned eddies twice [9, 10], NeuralPOM reduces velocity spectral error primarily by improving mesoscale phase alignment rather than inflating eddy amplitude; combining flow-dependent diapycnal damping factors $c(x)$ with probabilistic ensemble objectives or scale-aware spectral losses offers a path to jointly improve eddy amplitude and phase. Second, the present forecast experiments are conducted in an idealized closed double-gyre basin with zero surface buoyancy fluxes and complete fine-grid reference states; the differentiable PyTorch POM dynamical frameworks, conservation-constrained coupling interface, three-stage training procedure, and density-normal decomposition transfer directly to realistic regional and global primitive-equation ocean forecasting systems trained against high-resolution ocean reanalyses or observations.

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Data and code availability

The NeuralPOM source code is available at https://github.com/ChiyodaMomo01/NeuralPOM. The dataset is available at https://www.modelscope.cn/datasets/chiyodamomo01/NeuralPOM.

References

[1] Baylor Fox-Kemper, Alistair Adcroft, Claus W. Böning, Eric P. Chassignet, Enrique Curchitser, Gokhan Danabasoglu, Carsten Eden, Matthew H. England, Rüdiger Gerdes, Richard J. Greatbatch, Stephen M. Griffies, Robert W. Hallberg, Emmanuel Hanert, Patrick Heimbach, Helene T. Hewitt, Christopher N. Hill, Yoshiki Komuro, Sonya Legg, Julien Le Sommer, Simona Masina, Simon J. Marsland, Stephen G. Penny, Fangli Qiao, Todd D. Ringler, Anne Marie Treguier, Hiroyuki Tsujino, Petteri Uotila, and Stephen G. Yeager. Challenges and Prospects in Ocean Circulation Models. *Frontiers in Marine Science*, 6:65, 2019. doi: 10.3389/fmars.2019.00065. URL https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2019.00065/full.

[2] Patrick Heimbach, Fearghal O'Donncha, Timothy A. Smith, Jose Maria Garcia-Valdecasas, Alain Arnaud, and Liying Wan. Crafting the future: Machine learning for ocean forecasting. *State of the Planet*, 5-opsr:22, 2025. doi: 10.5194/sp-5-opsr-22-2025. URL https://sp.copernicus.org/articles/5-opsr/22/2025/.

[3] Helene T. Hewitt, Malcolm Roberts, Pierre Mathiot, Arne Biastoch, Ed Blockley, Eric P. Chassignet, Baylor Fox-Kemper, Pat Hyder, David P. Marshall, Ekaterina Popova, Anne-Marie Treguier, Laure Zanna, Andrew Yool, Yongqiang Yu, Rebecca Beadling, Mike Bell, Till Kuhlbrodt, Thomas Arsouze, Alessio Bellucci, Fred Castruccio, Bolan Gan, Dian Putrasahan, Christopher D. Roberts, Luke Van Roekel, and Qiuying Zhang. Resolving and Parameterising the Ocean Mesoscale in Earth System Models. *Current Climate Change Reports*, 6(4):137–152, 2020. doi: 10.1007/s40641-020-00164-w. URL https://link.springer.com/article/10.1007/s40641-020-00164-w.

[4] Yingzhe Cui, Ruohan Wu, Xiang Zhang, Ziqi Zhu, Bo Liu, Jun Shi, Junshi Chen, Hailong Liu, Shenghui Zhou, Liang Su, Zhao Jing, Hong An, and Lixin Wu. Forecasting the eddying ocean with a deep neural network. *Nature Communications*, 16(1):2268, 2025. doi: 10.1038/s41467-025-57389-2. URL https://www.nature.com/articles/s41467-025-57389-2.

[5] Anass El Aouni, Quentin Gaudel, Charly Regnier, Simon Van Gennip, Olivier Le Galloudec, Marie Drevillon, Yann Drillet, and Jean-Michel Lellouche. GLONET: Mercator's end-to-end neural global ocean forecasting system. *Journal of Geophysical Research: Machine Learning and Computation*, 2(3): e2025JH000686, 2025. doi: 10.1029/2025JH000686. URL https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JH000686.

[6] Daniel Holmberg, Emanuela Clementi, Italo Epicoco, and Teemu Roos. Accurate mediterranean sea forecasting via graph-based deep learning. *Scientific Reports*, 15(1):45051, 2025. doi: 10.1038/s41598-025-31177-w. URL https://www.nature.com/articles/s41598-025-31177-w.

[7] Yuan Niu, Qiusheng Huang, Xiaohui Zhong, Anboyu Guo, Lei Chen, Dianjun Zhang, Zhen-Qiang Zhou, Xiaoyan Jia, Libo Wu, Renhe Zhang, Jiawei Qi, Xuefeng Zhang, and Hao Li. A deep learning

global ocean forecasting model with sub-daily and eddy-resolving resolution. *npj Climate and Atmospheric Science*, 9(1):194, 2026. doi: 10.1038/s41612-026-01444-2. URL https://doi.org/10.1038/s41612-026-01444-2.

[8] Ashesh Chattopadhyay, Michael Gray, Tianning Wu, Anna B. Lowe, and Ruoying He. OceanNet: a principled neural operator-based digital twin for regional oceans. *Scientific Reports*, 14(1):21181, 2024. doi: 10.1038/s41598-024-72145-0. URL https://www.nature.com/articles/s41598-024-72145-0.

[9] C. E. Leith. Theoretical skill of Monte Carlo forecasts. *Monthly Weather Review*, 102(6):409–418, 1974. doi: 10.1175/1520-0493(1974)102<0409:TSOMCF>2.0.CO;2.

[10] Dmitrii Kochkov, Janni Yuval, Ian Langmore, Peter Norgaard, Jamie Smith, Griffin Mooers, Milan Klöwer, James Lottes, Stephan Rasp, Peter Düben, Sam Hatfield, Peter Battaglia, Alvaro Sanchez-Gonzalez, Matthew Willson, Michael P. Brenner, and Stephan Hoyer. Neural general circulation models for weather and climate. *Nature*, 632(8027):1060–1066, 2024. doi: 10.1038/s41586-024-07744-y. URL https://doi.org/10.1038/s41586-024-07744-y.

[11] Laure Zanna, William Gregory, Pavel Perezhogin, Aakash Sane, Cheng Zhang, Alistair Adcroft, Mitch Bushuk, Carlos Fernandez-Granda, Brandon Reichl, Dhruv Balwada, Julius Busecke, William Chapman, Alex Connolly, Danni Du, Kelsey Everard, Fabrizio Falasca, Renaud Falga, David Kamm, Etienne Meunier, Qi Liu, Antoine Nasser, Matthew Pudig, Andrew Shao, Julia L. Simpson, Linus Vogt, and Jiarong Wu. A framework for hybrid physics-AI coupled ocean models, 2025. URL https://arxiv.org/abs/2510.22676. Preprint; version 1 consulted.

[12] Thomas Bolton and Laure Zanna. Applications of Deep Learning to Ocean Data Inference and Subgrid Parameterization. *Journal of Advances in Modeling Earth Systems*, 11(1):376–399, 2019. doi: 10.1029/2018MS001472. URL https://doi.org/10.1029/2018MS001472.

[13] Laure Zanna and Thomas Bolton. Data-Driven Equation Discovery of Ocean Mesoscale Closures. *Geophysical Research Letters*, 47(17):e2020GL088376, 2020. doi: 10.1029/2020GL088376. URL https://doi.org/10.1029/2020GL088376.

[14] Arthur P. Guillaumin and Laure Zanna. Stochastic-Deep Learning Parameterization of Ocean Momentum Forcing. *Journal of Advances in Modeling Earth Systems*, 13(9):e2021MS002534, 2021. doi: 10.1029/2021MS002534. URL https://doi.org/10.1029/2021MS002534.

[15] Andrew Ross, Ziwei Li, Pavel Perezhogin, Carlos Fernandez-Granda, and Laure Zanna. Benchmarking of Machine Learning Ocean Subgrid Parameterizations in an Idealized Model. *Journal of Advances in Modeling Earth Systems*, 15(1):e2022MS003258, 2023. doi: 10.1029/2022MS003258. URL https://doi.org/10.1029/2022MS003258.

[16] Pavel Perezhogin, Cheng Zhang, Alistair Adcroft, Carlos Fernandez-Granda, and Laure Zanna. A Stable Implementation of a Data-Driven Scale-Aware Mesoscale Parameterization. *Journal of Advances in Modeling Earth Systems*, 16(10):e2023MS004104, 2024. doi: 10.1029/2023MS004104. URL https://doi.org/10.1029/2023MS004104.

[17] David Kamm, Pavel Perezhogin, Julie Deshayes, Etienne Meunier, and Alexis Barge. Assessing Data-Driven Eddy-Parameterizations in an Atlantic Sector Model. *Journal of Advances in Modeling Earth Systems*, 18(6):e2026MS005816, 2026. doi: 10.1029/2026MS005816. URL https://doi.org/10.1029/2026MS005816.

[18] Ruiqi Shu, Hao Wu, Yuan Gao, Fanghua Xu, Ruijian Gou, Wei Xiong, and Xiaomeng Huang. Advanced forecasts of global extreme marine heatwaves through a physics-guided data-driven approach. *Environmental Research Letters*, 20(4):044030, 2025. doi: 10.1088/1748-9326/adbddd. URL https://doi.org/10.1088/1748-9326/adbddd.

[19] Ruiqi Shu, Ruijian Gou, Yanfei Xiang, and Xiaomeng Huang. Ocean-E2E: Hybrid Physics-Based and Data-Driven Global Forecasting of Marine Heatwaves with End-to-End Neural Assimilation. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, pages 12018–12029, 2026. doi: 10.1145/3770855.3818893. URL https://doi.org/10.1145/3770855.3818893.

[20] Ruiqi Shu, Xiaohui Zhong, Qiusheng Huang, Ruijian Gou, Tianrun Gao, Hao Li, and Xiaomeng Huang. HybridOM: Hybrid physics-based and data-driven global ocean modeling with efficient spatial downscaling. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 112816–112846. PMLR, 2026. URL https://proceedings.mlr.press/v306/shu26b.html.

[21] Said Ouala, Bertrand Chapron, Fabrice Collard, Lucile Gaultier, and Ronan Fablet. Online calibration of deep learning sub-models for hybrid numerical modeling systems. *Communications Physics*, 7(1):402, 2024. doi: 10.1038/s42005-024-01880-7. URL https://doi.org/10.1038/s42005-024-01880-7.

[22] Hugo Frezat, Julien Le Sommer, Ronan Fablet, Guillaume Balarac, and Redouane Lguensat. A Posteriori Learning for Quasi-Geostrophic Turbulence Parametrization. *Journal of Advances in Modeling Earth Systems*, 14(11):e2022MS003124, 2022. doi: 10.1029/2022MS003124. URL https://doi.org/10.1029/2022MS003124.

[23] Bjoern List, Li-Wei Chen, Kartik Bali, and Nils Thuerey. Differentiability in unrolled training of neural physics simulators on transient dynamics. *Computer Methods in Applied Mechanics and Engineering*, 433:117441, 2025. doi: 10.1016/j.cma.2024.117441. URL https://doi.org/10.1016/j.cma.2024.117441.

[24] F. E. Yan, H. Frezat, J. Le Sommer, J. Mak, and K. Otness. Adjoint-Based Online Learning of Two-Layer Quasi-Geostrophic Baroclinic Turbulence. *Journal of Advances in Modeling Earth Systems*, 17(7):e2024MS004857, 2025. doi: 10.1029/2024MS004857. URL https://doi.org/10.1029/2024MS004857.

[25] George L. Mellor. An Equation of State for Numerical Models of Oceans and Estuaries. *Journal of Atmospheric and Oceanic Technology*, 8(4):609–611, 1991. doi: 10.1175/1520-0426(1991)008<0609:AEOSFN>2.0.CO;2. URL https://doi.org/10.1175/1520-0426(1991)008<0609:AEOSFN>2.0.CO;2.

[26] George L. Mellor and Tetsuji Yamada. Development of a turbulence closure model for geophysical fluid problems. *Reviews of Geophysics*, 20(4):851–875, 1982. doi: 10.1029/RG020i004p00851. URL https://doi.org/10.1029/RG020i004p00851.

[27] Alan F. Blumberg and George L. Mellor. A description of a three-dimensional coastal ocean circulation model. In Norman S. Heaps, editor, *Three-Dimensional Coastal Ocean Models*, volume 4 of *Coastal and Estuarine Sciences*, pages 1–16. American Geophysical Union, 1987. doi: 10.1029/CO004p0001. URL https://doi.org/10.1029/CO004p0001.

[28] S. Xu, X. Huang, L.-Y. Oey, F. Xu, H. Fu, Y. Zhang, and G. Yang. POM.gpu-v1.0: a GPU-based Princeton Ocean Model. *Geoscientific Model Development*, 8(9):2815–2827, 2015. doi: 10.5194/gmd-8-2815-2015. URL https://gmd.copernicus.org/articles/8/2815/2015/.

[29] Hao Wu, Yuan Gao, Ruiqi Shu, Kun Wang, Ruijian Gou, Chuhan Wu, Xinliang Liu, Juncai He, Shuhao Cao, Junfeng Fang, Xingjian Shi, Feng Tao, Qi Song, Shengxuan Ji, Yanfei Xiang, Yuze Sun, Jiahao Li, Fan Xu, Huanshuo Dong, Haixin Wang, Fan Zhang, Penghao Zhao, Xian Wu, Qingsong Wen, Deliang Chen, and Xiaomeng Huang. Advanced long-term earth system forecasting by learning the small-scale nature, 2025. URL https://arxiv.org/abs/2505.19432v1. Preprint; version 1 consulted, describing the historical Triton architecture.

[30] Zhangyang Gao, Cheng Tan, Lirong Wu, and Stan Z. Li. SimVP: Simpler yet better video prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3170–3180, 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Gao_SimVP_Simpler_Yet_Better_Video_Prediction_CVPR_2022_paper.html.

[31] Xingjian Shi, Zhourong Chen, Hao Wang, Dit-Yan Yeung, Wai-kin Wong, and Wang-chun Woo. Convolutional LSTM network: A machine learning approach for precipitation nowcasting. In *Advances in Neural Information Processing Systems*, volume 28, pages 802–810, 2015. URL https://papers.nips.cc/paper_files/paper/2015/hash/07563a3fe3bbe7e3ba84431ad9d055af-Abstract.html.

[32] Hao Wu, Fan Xu, Chong Chen, Xian-Sheng Hua, Xiao Luo, and Haixin Wang. PastNet: Introducing physical inductive biases for spatio-temporal video prediction. *arXiv preprint arXiv:2305.11421*, 2023. doi: 10.48550/arXiv.2305.11421. URL https://arxiv.org/abs/2305.11421.

[33] Stephen C. Bloom, Lawrence L. Takacs, Arlindo M. da Silva, and D. Ledvina. Data assimilation using incremental analysis updates. *Monthly Weather Review*, 124(6):1256–1271, 1996. doi: 10.1175/1520-0493(1996)124<1256:DAUIAU>2.0.CO;2.

[34] R. Penrose. A generalized inverse for matrices. *Proceedings of the Cambridge Philosophical Society*, 51(3):406–413, 1955. doi: 10.1017/S0305004100030401. URL https://www.cambridge.org/core/services/aop-cambridge-core/content/view/S0305004100030401.

[35] Alexander F. Shchepetkin and James C. McWilliams. A method for computing horizontal pressure-gradient force in an oceanic model with a nonaligned vertical coordinate. *Journal of Geophysical Research: Oceans*, 108(C3):3090, 2003. doi: 10.1029/2001JC001047. URL https://doi.org/10.1029/2001JC001047.

A Governing equations and numerical discretization

This appendix specifies the governing equations and discrete operators of NeuralPOM. Both dynamical frameworks solve the hydrostatic, Boussinesq primitive equations in terrain-following coordinates following the Princeton Ocean Model (POM) [27] and differ in their free-surface integration: the Explicit framework resolves the external barotropic mode through split-explicit subcycling, whereas the Implicit framework solves a semi-implicit elliptic free-surface system at the internal baroclinic time step. Validation against the double-precision Fortran reference code is presented in Appendix F.

A.1 Coordinates, variables, and time levels

Horizontal coordinates are x, y, the vertical coordinate z increases upward, and t denotes time. With resting bathymetric depth $h(x, y) > 0$ and free-surface elevation η, the total water-column thickness is $H = h + \eta$. Horizontal velocity components are u, v, physical vertical velocity is w_z, and g, ρ_0, f denote gravitational acceleration, reference seawater density, and the Coriolis parameter. Potential temperature T and salinity S are stored relative to reference offsets $(T_{\text{bias}}, S_{\text{bias}})$, and density ρ is represented by the normalized anomaly $r = (\rho - \rho_{\text{off}})/\rho_0$.

The terrain-following vertical coordinate $\sigma \in [-1, 0]$ and transformed vertical velocity Ω are defined by

$$\sigma = \frac{z - \eta}{H}, \qquad z = \eta + \sigma H, \qquad \Omega = H\frac{D\sigma}{Dt} = w_z - \partial_t\eta - u\partial_x\eta - v\partial_y\eta - \sigma(\partial_t H + u\partial_x H + v\partial_y H). \tag{A.1}$$

For any scalar field ϕ, horizontal and vertical derivatives transform according to

$$\partial_x\phi|_z = \partial_x\phi|_\sigma - \frac{\partial_x\eta + \sigma\partial_x H}{H}\partial_\sigma\phi, \qquad \partial_z\phi = \frac{1}{H}\partial_\sigma\phi, \tag{A.2}$$

and analogously in y. The water column is discretized into N_z active layers $k = 0, \ldots, N_z - 1$ bounded by interfaces $\sigma_0 = 0 > \sigma_1 > \cdots > \sigma_{N_z} = -1$, with layer thicknesses $\Delta\sigma_k = \sigma_k - \sigma_{k+1}$ $(\sum_{k=0}^{N_z-1} \Delta\sigma_k = 1)$, layer centers $\bar{\sigma}_k = \frac{1}{2}(\sigma_k + \sigma_{k+1})$, and center-to-center spacings $\Delta\bar{\sigma}_k = \bar{\sigma}_k - \bar{\sigma}_{k+1}$. An inactive bottom storage level $k = N_z$ is excluded from vertical integrals.

The internal baroclinic time step is Δt and the external barotropic substep is δt. Superscripts b, n, f denote the backward, current, and forward leapfrog time levels separated by $2\Delta t$ (internal) or $2\delta t$ (external), with free-surface pressure weight α and Robert–Asselin time-filter coefficient μ. The mode-split formulation tracks external elevation η_{E} $(H_{\text{E}} = h+\eta_{\text{E}})$, internal body elevation η_{B} $(H_{\text{B}} = h+\eta_{\text{B}})$, and the interval-averaged elevation η_{G} entering the internal momentum equations.

A.2 Hydrostatic momentum, continuity, and tracer equations

Let p denote hydrostatic pressure, $\eta_a = p_a/(\rho_0 g)$ atmospheric pressure head, K_M, K_H vertical turbulent viscosity and diffusivity, ν molecular background diffusivity, and $\mathcal{F}_u, \mathcal{F}_v$ horizontal viscous force divergences. In σ-coordinates, the conservative flux-divergence operator $\mathscr{C}(\phi)$ and volume continuity equation take the form

$$\mathscr{C}(\phi) = \partial_t(H\phi) + \partial_x(Hu\phi) + \partial_y(Hv\phi) + \partial_\sigma(\Omega\phi), \qquad \partial_t H + \partial_x(Hu) + \partial_y(Hv) + \partial_\sigma\Omega = 0. \tag{A.3}$$

Integrating continuity over $\sigma \in [-1, 0]$ with an impermeable bottom ($\Omega|_{\sigma=-1} = 0$), surface volume flux F_v, and depth-averaged velocities $(U, V) = \int_{-1}^{0} (u, v)\, d\sigma$ yields $\partial_t \eta + \nabla_h \cdot (HU, HV) + F_v = 0$. Writing $z_x = \partial_x \eta + \sigma \partial_x H$ and $z_y = \partial_y \eta + \sigma \partial_y H$, the horizontal momentum and hydrostatic relations are

$$\mathscr{C}(u) - Hfv = -\frac{H}{\rho_0}\left(\partial_x p - \frac{z_x}{H}\partial_\sigma p\right) + \partial_\sigma\left[\frac{K_M + \nu}{H}\partial_\sigma u\right] + H\mathcal{F}_u, \tag{A.4}$$

$$\mathscr{C}(v) + Hfu = -\frac{H}{\rho_0}\left(\partial_y p - \frac{z_y}{H}\partial_\sigma p\right) + \partial_\sigma\left[\frac{K_M + \nu}{H}\partial_\sigma v\right] + H\mathcal{F}_v, \qquad \partial_\sigma p = -gH\rho. \tag{A.5}$$

Potential temperature T and salinity S evolve under horizontal diffusion along σ-surfaces (acting on backward-level anomalies relative to reference climatology T_c, S_c with turbulent Prandtl ratio $A_H/A_M = 0.20$) and vertical turbulent diffusion:

$$\mathscr{C}(T) = \nabla_h \cdot \left[HA_H \nabla_h (T - T_c)\right] + \partial_\sigma\left[\frac{K_H + \nu}{H}\partial_\sigma T\right] + HQ_T, \qquad \mathscr{C}(S) = \nabla_h \cdot \left[HA_H \nabla_h (S - S_c)\right] + \partial_\sigma\left[\frac{K_H + \nu}{H}\partial_\sigma S\right] + HQ_S. \tag{A.6}$$

Vertical momentum boundary conditions prescribe surface wind stress $(K_M + \nu)\partial_z(u, v)|_{z=\eta} = \boldsymbol{\tau}_s/\rho_0$ and quadratic bottom drag $(K_M + \nu)\partial_z(u, v)|_{z=-h} = C_D|\boldsymbol{u}_b|\boldsymbol{u}_b$. In-situ density ρ and empirical sound speed c are evaluated from the nonlinear POM equation of state [25, 27] as functions of potential temperature $\vartheta = T + T_{\text{bias}}$ (°C), salinity $s = S + S_{\text{bias}}$, and static reference pressure $P = -\rho_0 g \bar{\sigma} h \times 10^{-5}$ (bar):

$$\rho = \rho(\vartheta, s, P), \qquad r = \frac{\rho(\vartheta, s, P) - \rho_{\text{off}}}{\rho_0}. \tag{A.7}$$

A.3 Arakawa-C spatial discretization and Smagorinsky viscosity

Horizontal operators are discretized on an orthogonal Arakawa-C grid where scalar cell (i, j) has grid spacings $\Delta x_{ij}, \Delta y_{ij}$, area $A_{ij} = \Delta x_{ij} \Delta y_{ij}$, and wet mask $m_{T,ij}$, while face velocities u_{ij} (between cells $i-1$ and i) and v_{ij} (between cells $j-1$ and j) carry control areas A^u_{ij}, A^v_{ij}, interpolated depths $H^u_{ij} = \frac{1}{2}(H_{ij} + H_{i-1,j})$ and $H^v_{ij} = \frac{1}{2}(H_{ij} + H_{i,j-1})$, transverse face lengths L^u_{ij}, L^v_{ij}, and face masks $m_{U,ij}, m_{V,ij}$. For masked face transports $Q^u = m_U H^u u$ and $Q^v = m_V H^v v$, the discrete horizontal flux divergence is

$$\left(D_h(Q^u, Q^v)\right)_{ij} = \frac{L^u_{i+1,j} Q^u_{i+1,j} - L^u_{ij} Q^u_{ij} + L^v_{i,j+1} Q^v_{i,j+1} - L^v_{ij} Q^v_{ij}}{A_{ij}}. \tag{A.8}$$

Second-order centered differences discretize advective momentum and tracer fluxes at time level n. Horizontal viscous and diffusive fluxes act on backward-level (b) velocity strain and climatological tracer anomalies $\phi'^b = \phi^b - \phi_c$ using resting face depths h^u, h^v, advancing tracers to the intermediate state $\phi^*_{ij,k} = (A_{ij} H^b_{ij} \phi^b_{ij,k} - 2\Delta t\, \mathcal{L}_{\phi,ij,k})/(A_{ij} H^f_{ij})$ prior to implicit vertical diffusion. Horizontal eddy viscosity follows the deformation-dependent Smagorinsky closure with cell-centered normal strain rates $D_x u, D_y v$ and interpolated shear s_{xy}:

$$A_M = C_s\, \Delta x\, \Delta y \sqrt{(D_x u)^2 + (D_y v)^2 + \tfrac{1}{2} s^2_{xy}}, \qquad C_s = 0.05. \tag{A.9}$$

A.4 Discrete hydrostatic baroclinic pressure gradient

Let $r' = r - r_{\text{mean}}$ denote the normalized density anomaly relative to the horizontally uniform reference profile r_{mean}. For the two scalar columns adjacent to a u-face, define $\Delta r'_k = r'_{i,k} - r'_{i-1,k}$ and $\Sigma r'_k = r'_{i,k} + r'_{i-1,k}$.

Vertical integration of the σ-transformed hydrostatic relation yields the discrete baroclinic pressure-gradient recurrence

$$B_0^u = \frac{g}{2}(-\bar{\sigma}_0)(H_i + H_{i-1})\Delta r'_0, \tag{A.10}$$

$$B_k^u = B_{k-1}^u + \frac{g}{4}(\bar{\sigma}_{k-1} - \bar{\sigma}_k)(H_i + H_{i-1})(\Delta r'_k + \Delta r'_{k-1}) + \frac{g}{4}(\bar{\sigma}_{k-1} + \bar{\sigma}_k)(H_i - H_{i-1})(\Sigma r'_k - \Sigma r'_{k-1}). \tag{A.11}$$

The corresponding horizontal baroclinic force is $\mathcal{P}_k^u = H^u L^u B_k^u m_U$ (and analogously $\mathcal{P}_k^v$ in y). Equations (A.10)–(A.11) enter both the internal momentum solver and the baroclinic pressure-gradient term of the training loss (Appendix B.4).

A.5 Vertical turbulence closure and implicit column diffusion

Vertical turbulent mixing is parameterized by the Mellor–Yamada level-2.5 closure [26] with prognostic equations for twice the turbulent kinetic energy q^2 and the master length-scale product $q^2 l$:

$$\mathscr{C}(q^2) = \nabla_h \cdot (HA_M \nabla_h q^2) + \partial_\sigma\Big[\frac{K_Q + \nu}{H}\partial_\sigma q^2\Big] + 2H\Big(P_s + B - \frac{q^3}{B_1 l}\Big), \tag{A.12}$$

$$\mathscr{C}(q^2 l) = \nabla_h \cdot \big[HA_M \nabla_h (q^2 l)\big] + \partial_\sigma\Big[\frac{K_Q + \nu}{H}\partial_\sigma (q^2 l)\Big] + H\Big(E_1 l(P_s + B) - \frac{q^3}{B_1} W_l\Big), \tag{A.13}$$

where $P_s = K_M[(\partial_z u)^2 + (\partial_z v)^2]$ is vertical shear production, $B = K_H b$ is buoyancy production, $W_l = 1 + E_2[l\kappa^{-1}((z+h)^{-1} + (\eta - z)^{-1})]^2$ is the wall-proximity function, and $(B_1, E_1, E_2, \kappa) = (16.6, 1.8, 1.33, 0.40)$. At interface k, with compressibility-corrected sound speed c_k from the POM equation of state, the vertical buoyancy gradient b_k, stability parameter G_H, and dissipation coefficient d_k are given by

$$b_k = \frac{g(r_{k-1} - r_k)}{\Delta\bar{\sigma}_{k-1} H} + \frac{2g^2}{c_{k-1}^2 + c_k^2}, \qquad G_H = \min\Big(\frac{l_k^2 b_k}{|q_b^2|}, 0.028\Big), \qquad d_k = \frac{\sqrt{|q_b^2|}}{B_1 l_k + \epsilon_l}. \tag{A.14}$$

With closure constants $(A_1, A_2, B_2, C_1) = (0.92, 0.74, 10.1, 0.08)$, the stability functions $S_H(G_H), S_M(G_H)$ determine the updated vertical diffusivities K_M, K_H, K_Q, which are averaged equally with their values from the preceding step. Boundary conditions for q^2 and $q^2 l$ are set by surface and bottom kinematic stress magnitudes a_s, a_b:

$$l_0 = 2 \times 10^5 a_s / g, \qquad Q_s = (15.8 \times 100)^{2/3} a_s, \qquad Q_{\text{bot}} = B_1^{2/3} a_b, \tag{A.15}$$

$$L_1 = -\kappa\sigma_1 H_f q_1^2, \qquad L_{N_z - 1} = \kappa(1 + \sigma_{N_z - 1}) H_f q_{N_z - 1}^2. \tag{A.16}$$

Vertical diffusive updates for $(u, v, T, S, q^2, q^2 l)$ are solved implicitly over $2\Delta t$ via tridiagonal column inversions across the active layers, with quadratic bottom drag $\gamma_b = C_D |\boldsymbol{u}_b^b|$ entering the bottom momentum diagonal and logarithmic bottom drag coefficient

$$C_D = \text{clip}\left[\left\{\frac{\kappa}{\ln[(z_{\text{sh}} + (1 + \bar{\sigma}_{\text{bottom}})H_{\text{E}})/z_0]}\right\}^2,\ C_{D,\min},\ C_{D,\max}\right], \tag{A.17}$$

where $C_{D,\min} = 0.0025$ and $C_{D,\max} = 1$.

A.6 Explicit mode-split integration

With mode-split step ratio $N_e = \Delta t/\delta t$, the Explicit framework advances the external barotropic mode over $m = 0, \dots, N_e - 1$ substeps driven by depth-averaged baroclinic forcing held fixed across the internal step:

$$\eta^{m+1} = \eta^{m-1} - 2\delta t\left[D_h(H^{u,m}U^m, H^{v,m}V^m) + F_v^m\right], \tag{A.18}$$

$$H^{u,m+1}U^{m+1} = H^{u,m-1}U^{m-1} - \frac{2\delta t}{A^u}\left[\overline{\mathcal{A}}^u - A^uC^u + gH^uL^u\delta_x\chi + \overline{\mathcal{P}}^u + A^u(a_s^u - a_b^u)\right], \tag{A.19}$$

where $\chi = (1-2\alpha)\eta^m + \alpha(\eta^{m-1} + \eta^{m+1}) + \eta_a$, C^u is the depth-weighted Coriolis term, and V^{m+1} is updated analogously. At each step, leapfrog levels are damped by the Robert–Asselin filter $\widetilde{x}^n = x^n + \frac{\mu}{2}(x^b - 2x^n + x^f)$ (with the vertical mean removed from $\Delta^2 u_k, \Delta^2 v_k$ so that three-dimensional filtering preserves the barotropic transport). Across the N_e substeps, the solver accumulates the forward body elevation $\eta_{\mathrm{B}}^f = \frac{\mu}{4}\eta_f^{(N_e-2)} + \frac{1}{2}(1 - \frac{\mu}{2})\eta_f^{(N_e-1)} + \frac{1}{2}\eta_f^{(N_e)}$, the interval-averaged elevation η_{G}^f, and the mean external transports $Q_+^{u,v} = N_e^{-1}\sum_{m=0}^{N_e-1} H^{u,v,(m)}(U,V)^{(m)}$.

A.7 Semi-implicit free-surface solver and implicit adjoint

The Implicit framework advances the external mode directly over Δt. Forming predictor transports $Q_*^u = H^{u,n}U_* = H^{u,b}U^b - (2\Delta t/A^u)\mathcal{N}_*^u$ (and Q_*^v analogously) with explicit pressure head $\chi_* = (1-2\alpha)\eta^n + \alpha\eta^b + \eta_a$ and right-hand side $b_\eta = \eta^b - 2\Delta t[D_h(Q_*^u, Q_*^v) + F_v^f]$, the forward free-surface elevation η^f satisfies the elliptic system

$$L_H\phi = D_h(m_U H^u G_x\phi,\ m_V H^v G_y\phi), \qquad A_\eta = I - 4\Delta t^2 g\alpha L_{H^n}, \qquad A_\eta\eta^f = b_\eta. \tag{A.20}$$

The velocity corrector updates the barotropic transport via $H^{u,f}U^f = H^{u,n}U_* - 2\Delta t\, g\alpha H^{u,f}G_x\eta^f$ in the double-gyre forecast experiments and via the conservative form

$$H^{u,f}U^f = H^{u,n}U_* - 2\Delta t\, g\alpha H^{u,n}G_x\eta^f \tag{A.21}$$

in the Fortran validation suite. The linear system $A_\eta\eta^f = b_\eta$ is solved by preconditioned conjugate gradients (PCG) with a fast discrete sine transform (DST-I) spectral preconditioner $\mathcal{S}$ based on the constant-coefficient Helmholtz eigenvalues $\lambda_{ab} = 4\Delta x_0^{-2}\sin^2[\pi a/(2(n_x+1))] + 4\Delta y_0^{-2}\sin^2[\pi b/(2(n_y+1))]$ evaluated at the median resting depth H_0,

$$B^{-1}r = \mathcal{S}^{-1}\left[\frac{\mathcal{S}(r)}{1 + 4\Delta t^2 g\alpha H_0\lambda_{ab}}\right]. \tag{A.22}$$

To propagate sensitivities through the elliptic solve without unrolling PCG iterations, we apply implicit differentiation. For any state or parameter ζ entering A_η or b_η and scalar objective $\mathcal{L}$ with upstream gradient $g_\eta = \partial\mathcal{L}/\partial\eta^f$, differentiating $A_\eta\eta^f = b_\eta$ gives

$$A_\eta^\top\lambda = g_\eta, \qquad \frac{d\mathcal{L}}{d\zeta} = \lambda^\top\left[\frac{\partial b_\eta}{\partial\zeta} - \frac{\partial A_\eta}{\partial\zeta}\eta^f\right]. \tag{A.23}$$

Because A_η is self-adjoint with respect to the cell-area inner product $M_A = \mathrm{diag}(A_{ij})$, $A_\eta^\top = M_A A_\eta M_A^{-1}$ and the adjoint variable is obtained from a single PCG solve:

$$\lambda = M_A A_\eta^{-1}(M_A^{-1}g_\eta). \tag{A.24}$$

A.8 Projection and three-dimensional mode coupling

Prior to internal baroclinic transport, horizontal velocities are constrained to match the two-interval average external transport $\overline{Q}^{u,v} = \frac{1}{2}(Q_-^{u,v} + Q_+^{u,v})$ while preserving vertical shear, $u_k \leftarrow u_k - \sum_\ell \Delta\sigma_\ell u_\ell + \overline{Q}^u/H_{\mathrm{B}}^u$

(and analogously for v_k). The transformed vertical velocity Ω is then diagnosed downward from the surface condition $\Omega_0 = \overline{F}_v$ via discrete continuity:

$$\Omega_{k+1} = \Omega_k + \Delta\sigma_k \left[D_h(H_{\mathrm{B}}^u u_k, H_{\mathrm{B}}^v v_k) + \frac{\eta_{\mathrm{B}}^f - \eta_{\mathrm{B}}^b}{2\Delta t} \right]. \tag{A.25}$$

Three-dimensional horizontal momentum is subsequently advanced with surface pressure gradient $\delta_x \chi_3$ ($\chi_3 = \frac{1}{2}(\eta_{\mathrm{G}}^f + \eta_{\mathrm{G}}^b) + \eta_a$) and baroclinic pressure-gradient force $\mathcal{P}_k^{u,v}$, followed by the implicit vertical viscosity solve, the vertical-mean-preserving Robert–Asselin filter, and the leapfrog time-level shift.

B State Correction Interface, Objective, and Training

B.1 Data clock and target alignment

Each forecast initialized at start day d begins from the fine-grid reference state projected onto the coarse grid at the end of day $d - 1$. Daily reference targets correspond to day-boundary states; hourly reference targets are extracted from the 20-minute reference fields archived during the first four hours of each day, so that each one-hour correction interval spans six 600-s coarse-grid internal steps (three 20-minute reference intervals). An offline cache stores the uncorrected coarse-grid state after the first physical integration interval, providing the exact pre-correction state for single-step pretraining and for the first coupling step of multi-step differentiable training.

B.2 Neural corrector architecture

The network input concatenates the three-dimensional current and backward leapfrog horizontal velocities and tracers (u, u^b, v, v^b, T, S) across all 16 stored vertical levels together with five two-dimensional external-mode fields (free-surface elevation and current and backward depth-averaged velocities), yielding $6 \times 16 + 5 = 101$ input channels, each standardized by frozen training-set mean and standard deviation statistics. The architecture is a two-dimensional residual U-Net with base width 256 and three 2×2 max-pooling stages (512, 1024, and 2048 channels), paired with a bilinear-upsampling decoder and skip connections. Each residual block consists of two bias-free 3×3 convolutions with group normalization and ReLU activations, supplemented by a 1×1 projection shortcut when channel width changes. The final 1×1 output convolution is initialized to zero so that training begins from an exact zero-increment identity mapping. The network contains 156×10^6 parameters and executes in 32-bit floating point with activation checkpointing during multi-step training.

B.3 Increment constraints and state update

The 60 output channels (four three-dimensional fields across the 15 active vertical layers) are scaled to physical units via $\delta f = s_f^{\mathrm{out}} a_{\theta,f}$ for $f \in \{u, v, T, S\}$. With horizontal cell area A_{ij}, scalar wet mask m_{ij}, resting depth h_{ij}, predicted body elevation $\eta_{\mathrm{B},ij}$, and layer thickness $\Delta\sigma_k$, the cell-volume weights are $w_{ijk} = A_{ij} m_{ij} (h_{ij} + \eta_{\mathrm{B},ij}) \Delta\sigma_k$. At every correction step, tracer increments are projected to zero layerwise volume mean and horizontal velocity increments are projected to zero column mean:

$$\delta T'_{ijk} = \delta T_{ijk} - \frac{\sum_{ij} w_{ijk} \delta T_{ijk}}{\sum_{ij} w_{ijk}}, \qquad \delta u'_{ijk} = m_{U,ij} \left[\delta u_{ijk} - \frac{\sum_k \Delta\sigma_k \delta u_{ijk}}{\sum_k \Delta\sigma_k} \right], \tag{B.1}$$

and analogously for S and v. When Density-Normal Damping (DND) is applied, the layer-mean-preserving orthogonal decomposition and density-normal attenuation $z_c = (I - E)z + cEz$ ($c = 3/4$, Appendix C) act on the standardized layer-mean-preserving tracer increment pair $(\delta T'/s_T, \delta S'/s_S)$. Each variable pair is then bounded by a domain-wide scalar limiter $c_{TS} = z_{\max}/\max(z_{\max}, p_{TS})$, where p_{TS} is the maximum absolute standardized tracer increment across both tracers and all active wet cells, and c_{uv} analogously for horizontal velocity, with $z_{\max} = 5$. In the state update, the resulting physical increments are added identically to both the current and backward leapfrog time levels, $f^n \leftarrow f^n + \widehat{\delta f}$ and $f^b \leftarrow f^b + \widehat{\delta f}$, the inactive bottom level $k = 15$ is left unchanged, and in-situ density ρ is updated across all wet cells via the nonlinear equation of state.

B.4 Composite endpoint loss

For variable f with predicted-geometry volume weights w^f and error normalization scale s_f^{err}, define the normalized mean squared error $N_f = \sum w^f (f - f^\star)^2 / \left(\sum w^f \, (s_f^{\mathrm{err}})^2\right)$. The momentum term averages current and backward leapfrog velocity errors, $N_m = \frac{1}{4}(N_u + N_{u^b} + N_v + N_{v^b})$. The baroclinic pressure-gradient consistency term N_p applies the discrete hydrostatic pressure-gradient recurrence (Eqs. (A.10)–(A.11)) to the equation-of-state density error $r_e = r - r^\star$, converts the resulting horizontal baroclinic force error (P_x, P_y) into an equivalent single-step leapfrog velocity increment,

$$k^u_{ijk} = -\frac{4\Delta t \, P_{x,ijk}}{(H_{ij} + H_{i-1,j}) A^u_{ij}} \, m_{U,ij}, \tag{B.2}$$

and analogously for k^v, and penalizes $N_p = \frac{1}{2}[M(k^u) + M(k^v)]/(s_p^{\mathrm{err}})^2$. The horizontal frontal-gradient term N_F averages the normalized squared differences of temperature and salinity errors between adjacent wet cells in both horizontal grid directions. The composite endpoint loss is

$$\ell_\chi = 0.25N_m + 0.25N_T + 0.25N_S + 0.15N_p + 0.10N_F + 0.10\,\chi N_\eta, \tag{B.3}$$

where $\chi = 0$ during Pretrain and $\chi = 1$ during Midtrain and Posttrain. Because the state update does not modify free-surface elevation directly, N_η contributes a nonzero parameter gradient only during Midtrain, where earlier velocity and tracer increments influence subsequent sea-surface height through the differentiable primitive-equation solver.

B.5 Optimization schedule and selected checkpoints

The three-stage training procedure progresses sequentially through Pretrain, Midtrain, and Posttrain. Pretrain optimizes single-step intervals ($N = 1$) on cached physical forecasts; Midtrain unrolls $N = 4$ coupled correction-and-integration intervals end-to-end through the differentiable dynamical framework; and Posttrain integrates 10-day closed-loop trajectories, performing one parameter update at the final correction of each day while detaching the computational graph between days. All three stages use eight-GPU data-parallel optimization with AdamW (weight decay 10^{-4}) and global gradient norm clipping at 10. Pretrain samples initialization states every six days across the seven 6000-day training reference members (7,000 samples per epoch) and trains for 40 epochs with a cosine learning-rate schedule decaying from 5×10^{-5} to 5×10^{-6} after a three-epoch linear warmup. Midtrain and Posttrain use a fixed learning rate of 3×10^{-6}. Hourly Midtrain samples trajectories every 24 days for 10 epochs; daily Midtrain samples every 72 days for three epochs. Posttrain samples 10-day rollouts every 60 days. Table 6 lists the completed optimizer updates for all four configurations; in the route ablations (Section 4.4), *Skip* omits Midtrain while *Detached* applies a stop-gradient operator at the entrance of each Midtrain physical interval $\mathcal{P}^\Delta$ under identical update budgets.

Table 6. Optimizer updates across training stages. Pretrain entries report total completed updates and the checkpoint selected by validation skill; Midtrain and Posttrain use their final completed checkpoints.

Configuration	Pretrain	Midtrain	Posttrain
Implicit, 1 h	35,000 / 34,125	2,190	1,780
Implicit, 1 day	35,000 / 35,000	222	5,038
Explicit, 1 h	35,000 / 33,250	2,190	1,780
Explicit, 1 day	35,000 / 35,000	222	4,846

C Layer-Constrained Density-Normal Projection

This appendix derives the layer-mean-preserving orthogonal projection used in Density-Normal Damping (DND, Section 5) and summarizes its numerical implementation.

C.1 Orthogonal projection onto the density-modifying subspace

Consider a single active σ-layer with wet cells i, normalized cell-volume weights $w_i > 0$ ($\sum_i w_i = 1$), and the Hilbert space of layer-mean-preserving standardized tracer increments $\mathcal{H}_0 = \{u = (u_i) : u_i \in \mathbb{R}^2, \ \sum_i w_i u_i = 0\}$ equipped with the volume-weighted inner product $\langle u, v\rangle_w = \sum_i w_i u_i^\top v_i$. Let $r_i = (\rho_i - \rho_{\text{off}})/\rho_0$ denote the normalized in-situ density anomaly at the pre-correction state, $g_i = (s_T \partial_T r_i, \, s_S \partial_S r_i)^\top$ its gradient in standardized tracer coordinates, $n_i = g_i/\|g_i\|_2$ the local unit density-normal direction, and $a_i = n_i^\top z_i$ the pointwise density-normal component of a given tracer increment $z \in \mathcal{H}_0$.

Because n_i varies horizontally across the basin due to the nonlinear equation of state, the pointwise density-normal field $a_i n_i$ does not in general have zero layer mean. The subspace of layer-mean-preserving density-modifying (diapycnal) increments is

$$\mathcal{S} = \left\{ p \in \mathcal{H}_0 : \ p_i = \beta_i n_i, \quad \sum_i w_i n_i \beta_i = 0 \right\}. \tag{C.1}$$

Finding the closest element $p \in \mathcal{S}$ to z in the $\|\cdot\|_w$ norm reduces to minimizing $\frac{1}{2}\sum_i w_i(\beta_i - a_i)^2$ subject to $\sum_i w_i n_i \beta_i = 0$. Introducing a Lagrange multiplier $\lambda \in \mathbb{R}^2$, the stationarity conditions give $\beta_i = a_i - n_i^\top \lambda$ and $M\lambda = b$, where $M = \sum_i w_i n_i n_i^\top \in \mathbb{R}^{2\times 2}$ is the symmetric positive semidefinite Gram matrix of the layer density normals and $b = \sum_i w_i n_i a_i \in \mathbb{R}^2$. Because any $t \in \ker M$ satisfies $0 = t^\top M t = \sum_i w_i (n_i^\top t)^2$ and hence $n_i^\top t = 0$ for all i, we have $t^\top b = 0$, so $b \in \operatorname{ran} M$. Applying the Moore–Penrose pseudoinverse M^+ [34] yields the unique minimum-norm multiplier $\lambda = M^+ b$ and the projected scalar amplitude

$$\beta_i = a_i - n_i^\top M^+ b, \qquad \sum_i w_i p_i = b - M M^+ b = 0. \tag{C.2}$$

For any test field $t_i = \gamma_i n_i \in \mathcal{S}$, $\langle z - p, t\rangle_w = \sum_i w_i (a_i - \beta_i)\gamma_i = \lambda^\top \sum_i w_i n_i \gamma_i = 0$. Thus $p = Ez$ defines the exact w-orthogonal projection of $\mathcal{H}_0$ onto $\mathcal{S}$, while $(I - E)z \in \mathcal{S}^\perp \cap \mathcal{H}_0$ is the layer-mean-preserving density-compensated (isopycnal) component.

Applying a directional retention factor $c \in [0, 1]$ to the density-modifying subspace produces the filtered increment

$$z_c = (I - E)z + cEz, \qquad \|z_c\|_w^2 = \|z\|_w^2 - (1 - c^2)\|Ez\|_w^2. \tag{C.3}$$

For DND ($c = 3/4$), Eq. (C.3) reduces the squared volume-weighted norm of the density-modifying component Ez by $1 - (3/4)^2 = 7/16 = 43.75\%$, while leaving the density-compensated component $(I - E)z$ and the layerwise heat and salt budgets unchanged.

C.2 Numerical implementation and norm-matched uniform control

At each correction step, the layer-mean-preserving tracer increment is computed in 64-bit precision and normalized by (s_T, s_S). Analytical partial derivatives $(\partial_T \rho, \partial_S \rho)$ of the POM equation of state (Eq. (A.7)) evaluated at the pre-correction state define the unit normal $n_i = g_i / \max(\|g_i\|_2, 10^{-30})$. Within each active layer, the 2×2 Gram matrix $M = \sum_i w_i n_i n_i^\top = \sum_{j=1}^{2} \lambda_j u_j u_j^\top$ is inverted via its truncated eigendecomposition,

$$M_\tau^+ = \sum_{\lambda_j > \tau \lambda_{\max}} \max(\lambda_j, 10^{-30})^{-1} u_j u_j^\top, \qquad \tau = 10^{-10}. \tag{C.4}$$

After forming $z_c = (I - E)z + cEz$ with $c = 3/4$ and applying the paired L_∞ limiter ($z_{\max} = 5$, Appendix B.3), the physical increments are added identically to (T^n, T^b) and (S^n, S^b), followed by a full nonlinear equation-of-state density update.

For the norm-matched Uniform control evaluated in Section 5.3, both tracer increments are scaled isotropically by $z_\kappa^{\text{uni}} = \kappa z$ with a fixed scalar factor κ calibrated once on cached Undamped development endpoints (days 1–10 from initialization days 8000, 8160, and 8320) to match the mean volume-weighted standardized tracer increment norm of DND: $\kappa = 0.9047$ for Implicit 1 h, 0.9064 for Implicit 1 day, 0.9313 for Explicit 1 h, and 0.9038 for Explicit 1 day.

D Evaluation Protocol and Physical Baselines

D.1 Error metrics

For forecast initialization origin $r \in \{1, \ldots, 16\}$ and lead day $d \in \{1, \ldots, 100\}$, three-dimensional prognostic fields are weighted by cell volume $w_{ijk}^{\text{ref}} = A_{ij} m_{ij}^f (h_{ij} + \eta_{\text{B},ij}^{\text{ref}}) \Delta\sigma_k$, using the fine-grid reference body elevation of that day and the variable's native staggered wet mask; sea-surface height uses the horizontal cell area $A_{ij} m_{ij}$. The spatial mean squared error $M_f(r, d)$ is evaluated across the 15 active vertical layers, and any forecast lead window $[a, b]$ of length $L = b - a + 1$ days is summarized by the pooled root-mean-square error and relative percentage change:

$$E_f(a, b) = \left[\frac{1}{16L} \sum_{r=1}^{16} \sum_{d=a}^{b} M_f(r, d) \right]^{1/2}, \qquad \Delta_f^{\text{ref}} = 100 \left[\frac{E_f}{E_f^{\text{ref}}} - 1 \right]. \tag{D.1}$$

All reductions are accumulated in 64-bit floating-point precision. Training objectives and layer-budget constraints use the predicted model geometry, whereas forecast evaluation uses the fine-grid reference geometry common to all methods at each origin and lead day.

D.2 Coarse-grid physical parameterizations

We compare three classical subgrid momentum and tracer closures on the coarse-grid POM dynamical frameworks:

- **Smagorinsky**: The standard second-order harmonic Smagorinsky horizontal viscosity $A_0 = c_m \, \Delta x \Delta y \, S_h$ and harmonic tracer diffusivity $A_H = c_c \, \Delta x \Delta y \, S_h$, where S_h is the resolved horizontal deformation rate. The reference coarse-grid baseline uses $c_m = 0.05$ and $c_c = 0.01$ (tprni = $c_c / c_m = 0.20$), and we additionally sweep independent (c_m, c_c) scaling pairs across both dynamical frameworks.

- **Biharmonic**: A scale-selective fourth-order horizontal momentum dissipation closure that replaces harmonic momentum viscosity with a symmetric fourth-order dissipative update at each internal step,

$$\boldsymbol{v}^{+} - \boldsymbol{v}^{-} = -\Delta t\, L^{\dagger_W} \kappa_4 L\, \tfrac{1}{2}(\boldsymbol{v}^{+} + \boldsymbol{v}^{-}), \qquad \kappa_4 = \gamma \frac{c_m}{0.05} \frac{A_0}{4(\Delta x^{-2} + \Delta y^{-2})}, \tag{D.2}$$

  where L is the discrete horizontal stress operator at unit viscosity, W is the diagonal matrix of wet-face control volumes, and $L^{\dagger_W} = W^{-1} L^{\top} W$ is its volume-weighted adjoint. The implicit midpoint update yields an exact single-step kinetic-energy dissipation rate of $\Delta t \langle L\bar{\boldsymbol{v}}, \kappa_4 L\bar{\boldsymbol{v}} \rangle_W$ with $\gamma = 2$.
- **Backscatter**: Fourth-order biharmonic viscosity coupled with a kinetic-energy backscatter parameterization. A fraction $\beta = 0.5$ of the biharmonic kinetic-energy loss is transferred to column subgrid energy reservoirs in regions where the flow is geostrophically balanced and the first baroclinic Rossby radius of deformation is resolved, and is reinjected into the resolved flow via anti-viscous harmonic forcing while accounting for the exact finite-step energy exchange ($\gamma = 2$). Across both dynamical frameworks, Biharmonic and Backscatter yield larger 100-day RMSE than harmonic Smagorinsky across all five prognostic variables.

D.3 Spatial tracer error decomposition and density

With normalized reference volume weights W_{ik} ($\sum_{i,k} W_{ik} = 1$), layer weights $W_k = \sum_i W_{ik}$, layer-mean error $\mu_k = W_k^{-1} \sum_i W_{ik} e_{ik}$, and global domain-mean error $\mu = \sum_k W_k \mu_k$, the three-dimensional tracer MSE decomposes orthogonally into domain-mean, vertical profile-shape, and horizontal anomaly components:

$$\sum_{i,k} W_{ik} e_{ik}^2 = \mu^2 + \sum_k W_k (\mu_k - \mu)^2 + \sum_{i,k} W_{ik} (e_{ik} - \mu_k)^2. \tag{D.3}$$

In-situ density errors are evaluated in $\mathrm{kg\,m^{-3}}$ from the nonlinear POM equation of state (Eq. (A.7)). Figure 6 shows the 100-day daily RMSE evolution of temperature and salinity across all four hybrid configurations and their coarse-grid Smagorinsky baselines.

D.4 Horizontal velocity spectra

Each horizontal velocity component is analyzed on its native staggered C-grid. Within each active layer k, the velocity field is masked, multiplied by a two-dimensional Hann window q_{ij}, demeaned with the same window weights, normalized by $\left[(N_x N_y)^{-1} \sum_{ij} q_{ij}^2\right]^{1/2}$, and transformed via forward-normalized two-dimensional fast Fourier transforms. Isotropic radial wavenumber bins span 0 to $\sqrt{1/2}$ cycles per grid interval. Layer kinetic energies $E_{kb} = \frac{1}{2} \sum_{\mathcal{B}_b} (|\hat{u}_k|^2 + |\hat{v}_k|^2)$ are weighted by layer thickness $\Delta\sigma_k$. Prediction, reference, error, and cross-spectral kinetic energies E^p, E^r, E^e, C are averaged across the 16 evaluation origins and lead days within each forecast window before forming the energy ratio $R = E^p/E^r$, normalized spectral error $Q = E^e/E^r$, and phase alignment $A = C/\sqrt{E^p E^r}$, which satisfy the exact identity $Q = R + 1 - 2A\sqrt{R}$. Spectral bands are partitioned into large scales ($k \le 0.125$ cycles per grid interval), intermediate scales ($0.125 < k \le 0.25$), near-grid scales ($0.25 < k \le 0.5$), and corner scales ($k > 0.5$).

D.5 Developed double-gyre circulation

Figure 7 shows an instantaneous snapshot of the fine-grid explicit reference simulation at model day 10,298 on the 352×320 wet interior grid. Relative vorticity ζ/f is computed as the spherical horizontal curl of face-averaged velocities in the uppermost active layer.

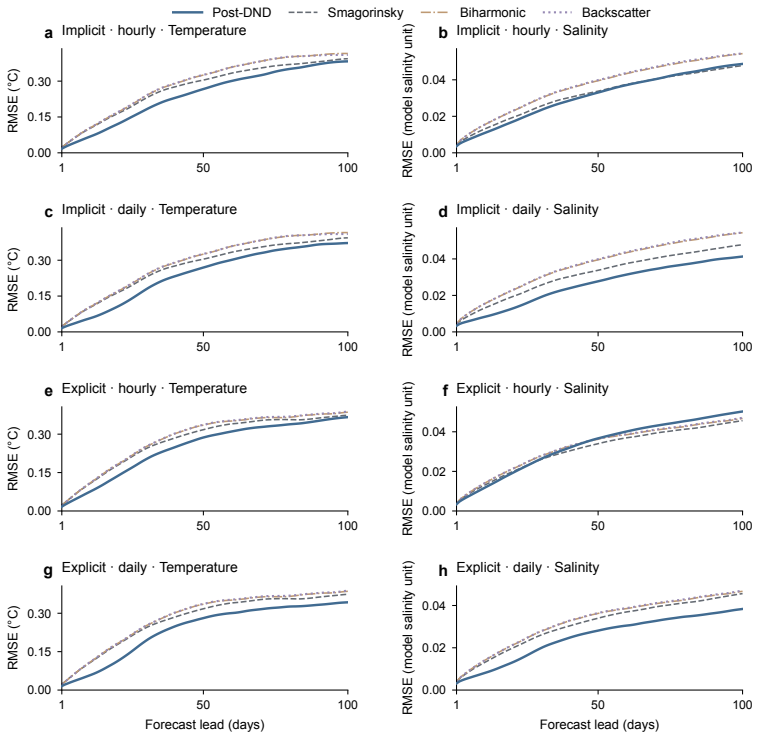


Figure 6. Tracer error evolution over 100 days. Daily potential temperature and salinity RMSE of the Posttrain DND hybrid forecasts and the coarse-grid Smagorinsky baselines, pooled across the 16 evaluation origins at each lead day.

D.6 Data archive and reproducibility

All hybrid, physical-baseline, and direct neural forecasts complete the full 100-day integration across all 16 evaluation origins. The accompanying `data/` archive provides the daily spatial MSE series for every trajectory, window-aggregated RMSE tables, three-component tracer MSE decompositions, vertical layer profiles, radially binned velocity spectra and band-integrated spectral metrics, the Undamped, Uniform, and DND comparisons, and the direct neural forecaster evaluations, together with FROZEN_PROTOCOL.json recording cryptographic hashes of the code, checkpoints, normalization constants, and reference datasets.

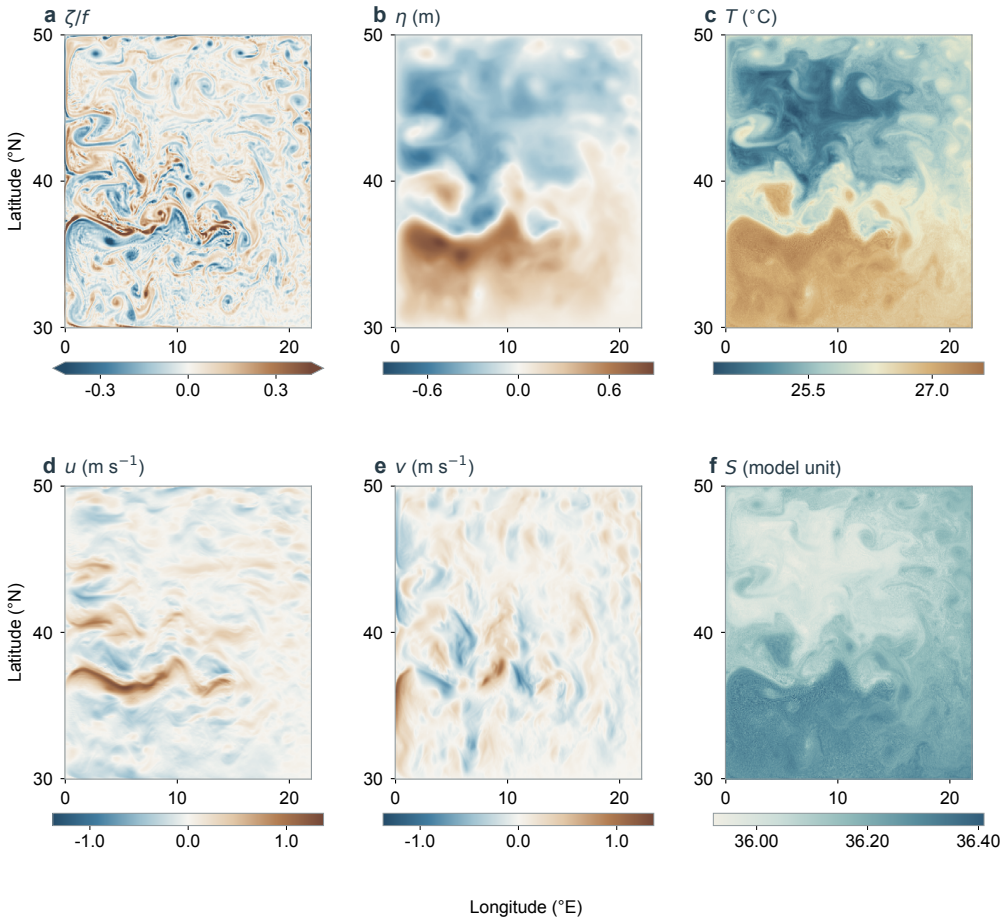


Figure 7. Developed circulation in the wind-driven double-gyre ocean basin. Uppermost-layer normalized relative vorticity ζ/f, free-surface elevation, potential temperature, zonal velocity, meridional velocity, and salinity of the fine-grid explicit reference integration at model day 10,298 on the 352×320 interior grid. Vorticity color limits saturate the largest 0.5% of absolute values.

E Direct Neural Forecasters

E.1 Full-state emulation task and architectures

The direct autoregressive neural forecasters are trained and evaluated on the semi-implicit reference dataset in the wind-driven double-gyre ocean basin using the same ensemble member and temporal split as NeuralPOM. At each daily step, each direct model takes as input and predicts the complete 313-channel model state: 18 three-dimensional variables across all 16 vertical levels (current and backward horizontal velocities and tracers, turbulent kinetic energy and mixing length variables, vertical momentum and tracer diffusivities, horizontal viscosity, in-situ density, and transformed vertical velocity Ω) together with 25 two-dimensional variables (free-surface elevations, depth-averaged velocities, interval transport accumulators, surface wind forcing, and bottom stress). Each channel is standardized using frozen training-set statistics; land and bottom cells are masked, and the model clock advances by one day per autoregressive step without invoking the primitive-equation solver.

We evaluate five spatiotemporal neural architectures (with validation-selected Pretrain and final completed Posttrain optimizer steps indicated in parentheses):

- **ResUNet** (Pretrain: 8,750; Posttrain: 11,440): The residual U-Net architecture of the hybrid corrector widened to 313 input and 313 output channels.
- **Triton** (Pretrain: 22,750; Posttrain: 17,327): The multi-scale spatiotemporal hierarchy of [29], configured with spatial hidden width 256, temporal hidden width 512, four spatial blocks, and eight temporal blocks.
- **SimVP** (Pretrain: 25,375; Posttrain: 7,920): The Inception-based spatiotemporal translator of [30] with spatial width 256 and temporal width 512.
- **ConvLSTM** (Pretrain: 34,125; Posttrain: 17,600): A two-layer convolutional LSTM network [31] with 256 hidden channels per layer and 3×3 kernels.
- **PastNet** (Pretrain: 9,625; Posttrain: 9,680): The hybrid spectral-convolution and discrete vector-quantized codebook architecture of [32] with a 512-entry codebook.

Horizontal grids are padded to 92×84 (Triton, SimVP) or 96×96 (PastNet) and cropped back to the 90×82 domain after each forward pass.

E.2 Two-stage training protocol and window evaluation

All five direct neural architectures are trained using the channel-averaged, area- and thickness-weighted mean squared error across all 313 standardized state channels:

$$\ell_{\mathrm{AI}} = \frac{1}{313} \sum_{c=1}^{313} \frac{\sum_i w_{ci} (\hat{a}_{ci} - a_{ci})^2}{\sum_i w_{ci}}. \tag{E.1}$$

During single-step pretraining (Pretrain), each network is trained on one-day state transitions sampled every six days across the seven 6000-day training members using the same optimizer and cosine learning-rate schedule as the hybrid Pretrain stage. During closed-loop rollout training (Posttrain), initialized from the validation-selected Pretrain checkpoint, each network is trained on 10-day autoregressive rollouts sampled every 60 days, performing one optimizer update per rollout day with the predicted state detached between consecutive daily steps.

Whereas Table 2 in Section 4.2 reports the full 100-day pooled RMSE across all 16 evaluation origins, Table 7 partitions performance into early (days 1–50) and late (days 51–100) forecast windows for all ten direct neural checkpoints, the coarse-grid Smagorinsky baseline, and the daily NeuralPOM hybrid forecast (DND).

Table 7. Early- and late-window forecast RMSE (days 1–50 and days 51–100). Shading and boldface denote the lowest RMSE in each column within each forecast window; underlining denotes the second lowest (see Table 2 for days 1–100). Horizontal velocities u, v in $\mathrm{m\,s^{-1}}$, sea-surface height (SSH) in m, potential temperature T in °C, and salinity S in model salinity units.

Configuration	$u\downarrow$	$v\downarrow$	SSH $\downarrow$	$T\downarrow$	$S\downarrow$
Days 1–50					
`Smagorinsky`	0.05715	0.06125	0.08291	0.20676	0.02327
Triton Pre	0.08046	0.08143	0.16975	0.60555	0.16690
Triton Post	0.06722	0.06694	0.11266	0.29419	0.09654
ResUNet Pre	0.08042	0.07820	0.12440	0.35888	0.09838
ResUNet Post	0.05796	0.05934	0.08642	0.43161	0.17765
SimVP Pre	0.08407	0.08155	0.19107	0.68022	0.18472
SimVP Post	0.06137	0.06152	0.10150	0.51095	0.22896
ConvLSTM Pre	0.10115	0.10333	0.30347	0.94709	0.21542
ConvLSTM Post	0.07765	0.07741	0.13209	0.39224	0.08424
PastNet Pre	0.11779	0.11206	0.18938	0.54899	0.18091
PastNet Post	0.08345	0.08456	0.13818	0.72325	0.26158
NeuralPOM Post-`DND` (ours)	**0.04813**	**0.05282**	**0.06935**	**0.16586**	**0.01769**
Days 51–100					
`Smagorinsky`	0.09644	0.09841	0.14759	0.35992	0.04179
Triton Pre	0.11518	0.10732	0.25301	0.94383	0.25964
Triton Post	0.10619	0.10261	0.17253	0.42704	0.12042
ResUNet Pre	0.11957	0.11386	0.19674	0.53868	0.14740
ResUNet Post	0.10154	0.10292	0.15925	0.74587	0.26585
SimVP Pre	0.10359	0.09997	0.25235	0.89031	0.24270
SimVP Post	0.10257	0.09798	0.16676	0.73448	0.28231
ConvLSTM Pre	0.10062	0.10370	0.28644	0.98514	0.23197
ConvLSTM Post	0.12194	0.11266	0.21862	0.61207	0.12609
PastNet Pre	0.16349	0.13989	0.26474	0.66849	0.17105
PastNet Post	0.11901	0.11599	0.19676	0.81440	0.27646
NeuralPOM Post-`DND` (ours)	**0.09317**	**0.09666**	**0.14283**	**0.33666**	**0.03577**

F Validation of the differentiable dynamical frameworks

Both differentiable PyTorch dynamical frameworks were validated in 64-bit floating point against the double-precision Fortran Princeton Ocean Model across four benchmark configurations spanning idealized basin, coastal, topographic, and realistic embayment regimes with identical initial states, bathymetry, forcing, and time steps:

1. **Double gyre**: A 24-hour spin-up of the wind-driven double-gyre ocean basin on a closed 90×82 horizontal grid with 15 active σ-layers, zero normal boundary velocities, internal step $\Delta t = 900\,\mathrm{s}$, external substep $\delta t = 30\,\mathrm{s}$, and $\mu = 0.10$ (whereas the long double-gyre forecast integrations in the main text use $\Delta t = 600\,\mathrm{s}$, $\delta t = 20\,\mathrm{s}$, $(\alpha, \mu) = (0.225, 0.10)$ for Explicit, and $(0.333, 0.15)$ for Implicit).
2. **Coastal tide**: A 15-day barotropic tidal integration on a 161×101 grid with four active layers, $\Delta t = 10\,\mathrm{s}$, $\delta t = 1\,\mathrm{s}$, fixed tracers, logarithmic bottom drag (Eq. (A.17) with $z_{\mathrm{sh}} = 0.01\,\mathrm{m}$ and $z_0 = 0.005\,\mathrm{m}$),

an eastern Flather radiation condition $U_E = U_{\mathrm{adj}} + \sqrt{g/H_{\mathrm{adj}}}(\eta_{\mathrm{adj}} - \eta_E)$, and western M_2 tidal forcing $\eta_{\mathrm{tide},j}(t_h) = a_j \cos(\omega t_h - \phi_j)$ enforced via Dirichlet boundary lifting in the Implicit solver.

3. **Stratified seamount**: A 10-day integration initialized from rest over an isolated seamount on a 65×49 grid with 15 active layers, $\Delta t = 150\,\mathrm{s}$, $\delta t = 5\,\mathrm{s}$, $r_{\mathrm{mean}} = 0$, open eastern and western radiation boundaries $U_{W,E} = \mp\sqrt{g/h}(\eta_{W,E} - \eta^{\mathrm{ext}}_{W,E})$ incorporated into A_η as diagonal Robin terms, and conservative body-elevation updates driven by $\overline{Q}$.
4. **Jiaozhou Bay**: A 15-day realistic-topography tidal integration of Jiaozhou Bay on a 312×213 horizontal grid ($\Delta x \approx 141.7\,\mathrm{m}$, $\Delta y \approx 185.0\,\mathrm{m}$, 30,159 interior wet water columns with $2.40\,\mathrm{m} \le h \le 60.03\,\mathrm{m}$) and 16 active σ-layers, with $\Delta t = 10\,\mathrm{s}$, $\delta t = 1\,\mathrm{s}$, fixed tracers, depth-gated logarithmic bottom drag ($z_{\mathrm{sh}} = 0.01\,\mathrm{m}$, $z_0 = 0.005\,\mathrm{m}$ evaluated where $H \ge h_c = 0.05\,\mathrm{m}$), gravity-wave and Orlanski velocity radiation boundaries, and M_2 tidal forcing across 389 active wet open-boundary nodes along the southern and eastern boundaries enforced via Dirichlet boundary lifting in the Implicit solver.

Across all four validation benchmarks, the Implicit solver uses the conservative transport corrector (Eq. (A.21)), area-weighted PCG inner products, and a relative residual tolerance of 10^{-11}.

The Explicit framework reproduces the Fortran reference to numerical round-off: after 24 hours in the double-gyre benchmark, its velocity RMSEs are 1.78×10^{-9} and $1.64 \times 10^{-9}\,\mathrm{m\,s^{-1}}$, and its sea-surface elevation RMSE is $8.51 \times 10^{-10}\,\mathrm{m}$ (Table 8 and Fig. 8). The Implicit framework replaces external mode subcycling with a semi-implicit elliptic solve at the baroclinic step; its 24-hour double-gyre elevation difference of 1.42 mm decreases to 0.46 mm when Δt is reduced from 900 s to 100 s, confirming temporal consistency between the two free-surface discretizations. In the coastal tidal benchmark, the root-mean-square elevation difference over hours 240–360 is 1.9 μm for Explicit and 0.50 mm for Implicit, with station M_2 tidal amplitudes and phases agreeing with Fortran to within 0.46 mm and 0.042°. In the resting stratified seamount benchmark, both PyTorch frameworks and Fortran POM develop the same terrain-following pressure-gradient error current of approximately $0.30\,\mathrm{m\,s^{-1}}$ after 10 days [35], with inter-model velocity differences below $10^{-4}\,\mathrm{m\,s^{-1}}$. In the realistic-topography Jiaozhou Bay benchmark, where an un-ramped cold-start M_2 tide propagates across complex coastal geometry and shallow shoals, the Explicit framework matches Fortran to machine precision over the initial steps ($\sim 10^{-16}\,\mathrm{m}$) and maintains a root-mean-square elevation difference of 0.094 mm over days 10–15 (0.19 mm at day 15), while the Implicit framework damps the initial six-hour cold-start gravity-wave transient at $\Delta t = 10\,\mathrm{s}$ before converging onto the forced tidal equilibrium with a root-mean-square elevation difference of 0.19 mm over days 10–15 (0.28 mm at day 15) and interior station M_2 amplitude and phase differences within 0.12 mm and 0.010° across both frameworks (Fig. 9).

Table 8. Root-mean-square differences against the double-precision Fortran POM reference at the end of each validation benchmark. Velocities u, v in $\mathrm{m\,s^{-1}}$, potential temperature T in °C, salinity S in model salinity units, and free-surface elevation η in m.

Benchmark	Framework	u	v	T	S	η
Double gyre (24 h)	Explicit	1.780e-09	1.638e-09	3.502e-09	1.981e-09	8.510e-10
Double gyre (24 h)	Implicit	7.357e-05	3.563e-05	2.303e-06	1.438e-05	1.416e-03
Coastal tide (15 d)	Explicit	1.349e-06	7.935e-07	0	0	1.224e-06
Coastal tide (15 d)	Implicit	7.175e-04	1.858e-04	0	0	5.466e-04
Seamount (10 d)	Explicit	6.298e-05	2.382e-05	1.764e-04	9.766e-08	4.439e-05
Seamount (10 d)	Implicit	9.304e-05	4.911e-05	1.973e-04	9.540e-08	3.002e-05
Jiaozhou Bay (15 d)	Explicit	2.916e-03	5.724e-03	0	0	1.866e-04
Jiaozhou Bay (15 d)	Implicit	5.371e-03	7.468e-03	0	0	2.840e-04

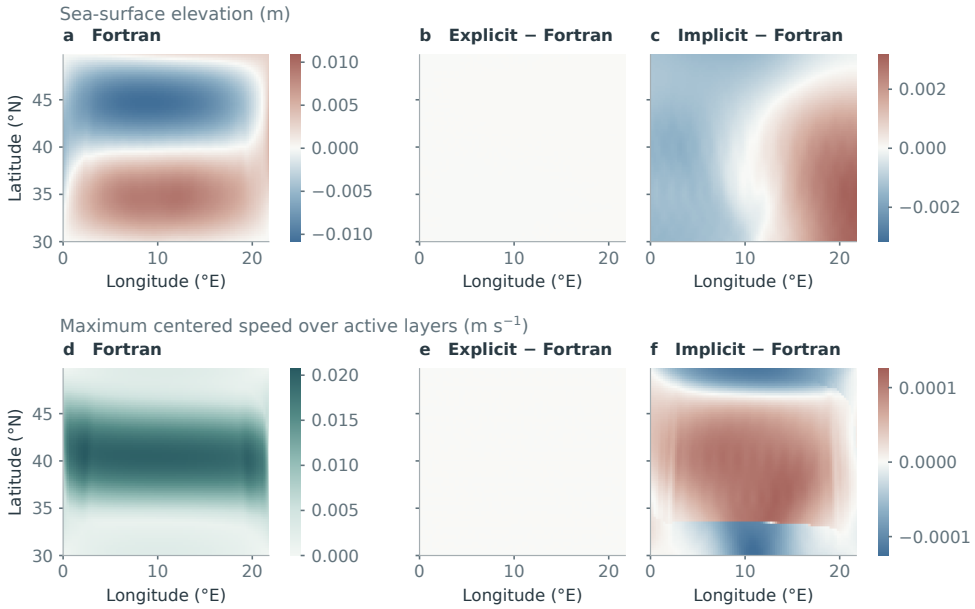


Figure 8. Double-gyre validation benchmark at 24 hours. Fortran POM reference fields (left) and pointwise differences for the Explicit (middle) and Implicit (right) differentiable PyTorch frameworks, shown for free-surface elevation (top) and column-maximum horizontal speed (bottom).

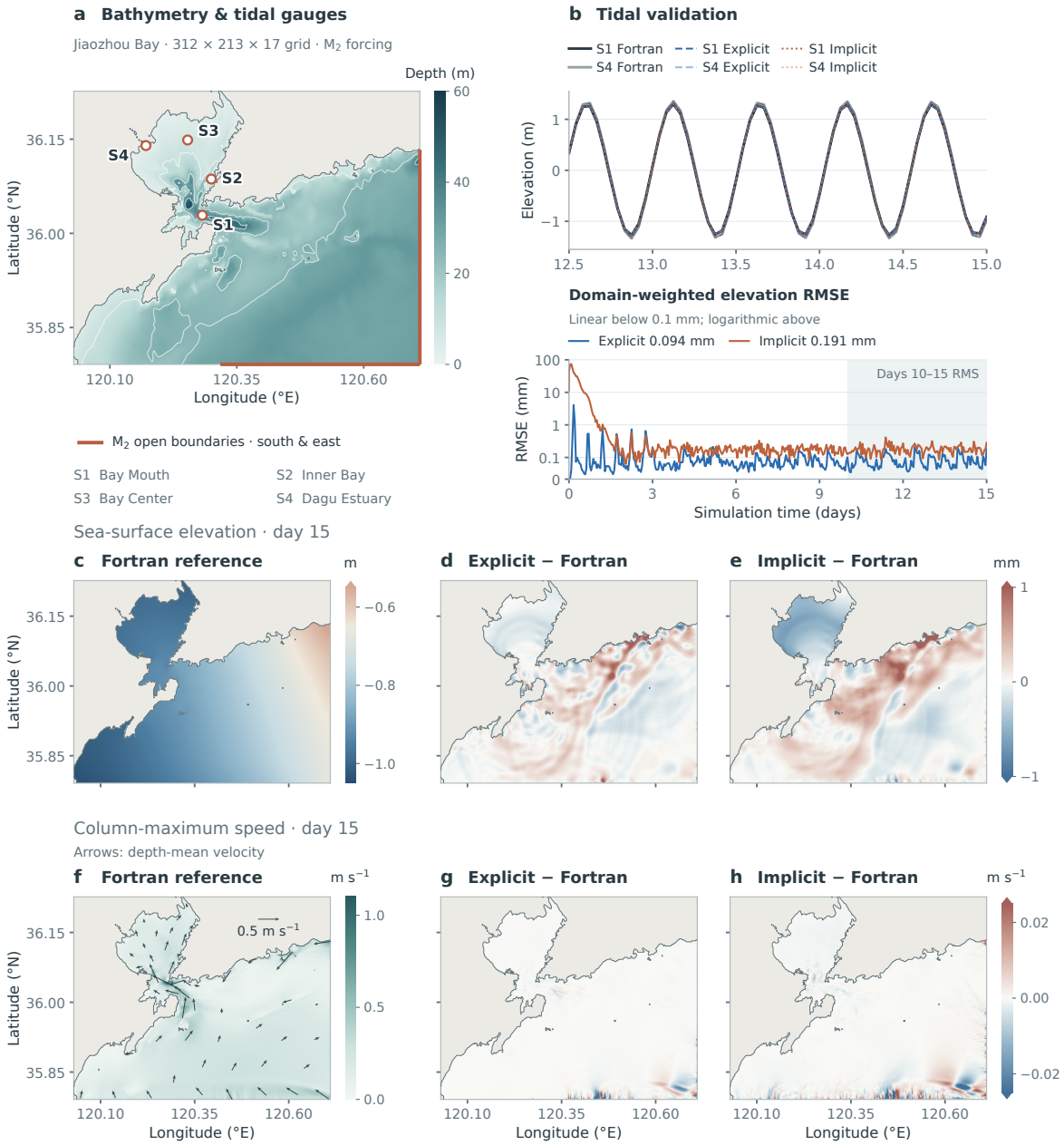


Figure 9. Realistic-topography Jiaozhou Bay validation benchmark over a 15-day (360 h) M_2 tidal integration. (**a**) Coastal bathymetry h, southern and eastern M_2 open boundaries, and interior tidal gauge locations (S1 Bay Mouth, S2 Inner Bay, S3 Bay Center, S4 Dagu Estuary). (**b**) Equilibrium tidal gauge elevation series over days 12.5–15 (top) and 15-day domain-weighted free-surface elevation RMSE time series against the double-precision Fortran POM reference (bottom). (**c–e**) Day-15 free-surface elevation η for the Fortran reference (m) and pointwise differences (mm) for the Explicit and Implicit differentiable PyTorch frameworks. (**f–h**) Day-15 column-maximum horizontal speed ($m\,s^{-1}$) with depth-averaged current vectors overlaid on the Fortran reference, alongside pointwise speed differences for the Explicit and Implicit frameworks.